



A COMPARATIVE STUDY OF TWO
METHODS FOR ESTIMATING CAUSAL
EFFECTS WITH TIME-VARYING
TREATMENTS UNDER
TIME-INVARIANT UNMEASURED
CONFOUNDING

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Declaration of Academic Integrity

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1 Introduction

Many fundamental questions in social sciences and epidemiology are causal in nature focussing on the *effect* of an *intervention* on an outcome. Examples include whether increasing the minimum wage affects employment (Card & Krueger, 1993) or whether initiating hormone therapy changes the risk of coronary heart disease in postmenopausal women (Hernán et al., 2008). In order to answer such questions, it is not sufficient to study associations alone. Rather, causal inference requires assumptions about the underlying data-generating process (DGP) and a clear definition of the causal quantity of interest (Greenland, Pearl, & Robins, 1999; Hernán & Robins, 2016). While randomized controlled trials are often regarded as the gold standard for estimating causal effects, they are usually infeasible, unethical, or expensive in practice. As a result, researchers often have to rely on observational data to estimate causal effects, which introduces additional challenges due to potential confounding.

This thesis focuses on a specific observational longitudinal study design with discrete time points (Gill & Robins, 2001), which involves data with repeated covariate measurements of the same units, a time-varying binary treatment, a terminal outcome at the end of the observation period and a lack of anticipation effects, meaning that treatments at future points do not impact potential outcomes at the current time point (Arkhangelsky & Imbens, 2024). Instances of such designs can be found in psychology (Molina et al., 2023), sociology (Bacak & Kennedy, 2015), and epidemiology (Bodnar, Davidian, Siega-Riz, & Tsiatis, 2004). Moreover, the growing availability of intensive longitudinal data through techniques like experience sampling, mobile evaluations, and wearable devices has made methods for longitudinal causal inference increasingly important (Myin-Germeys et al., 2018; Shi & Dempsey, 2025).

Estimating causal effects in longitudinal settings with time-varying treatments and confounders is particularly challenging due to the presence of treatment-confounder feedback, where treatments at previous time points can affect confounders at subsequent time points, which in turn can affect future treatments and outcome. In presence of such feedback loops, naive standard regression methods that adjust for post-treatment confounders at each time point can produce biased estimates of causal effects. To address this issue, a family of methods known as the g-methods have been developed (Robins & Hernán, 2008; Naimi, Cole, & Kennedy, 2017). However, these methods rely on the assumption of no unmeasured

confounding or sequential ignorability. This assumption states that all confounders of the treatment-outcome relationship are measured and included in the analysis, which is often violated in observational studies.

There are two types of unmeasured confounding that can occur in longitudinal studies: time-varying and time-invariant. In this thesis, the focus is on the latter, which are also referred to as missing baseline confounders. These are unmeasured covariates that affect the treatments and outcome at all-time points but do not change over time. Examples of such baseline covariates may include genetic predisposition, personality traits, or other stable characteristics of individuals during the span of the study.

Two methods have been proposed in the recent literature to estimate the causal effects of time-varying treatments in the presence of time-invariant unmeasured confounders. The first is the fixed-effect marginal structural model (FE-MSM) proposed by Blackwell and Yamauchi (2024), which combines the marginal structural model (MSM) (Robins, Hernán, & Brumback, 2000) with unit fixed effects. The second method is the sequential Gaussian process latent variable model (SeqGPLVM) introduced by Hatt and Feuerriegel (2024). This method may be interpreted as the sequential extension of the deconfounder theory (Wang & Blei, 2019a).

These two methods are comparable as they can be used in similar settings such as those described above. They both involve a unit-specific constant unobserved heterogeneity component that is supposed to capture the effect of the unmeasured confounders. They diverge, however, in their characterization of the unobserved heterogeneity component. Whereas FE-MSM—rooted in econometrics literature—conceptualizes this component as a frequentist fixed effect and a nuisance parameter which is tied to inverse probability of treatment weighting (IPTW) of marginal structural models, the machine-learning-based SeqGPLVM views it as a latent variable and employs a much more flexible and non-parametric approach to estimate that latent variable which then can be employed in a wide range of causal inference estimators. As a result, they make different assumptions about the unobserved heterogeneity component, the range of causal targets they can identify and the scope of estimators they can be used with. Despite these similarities and differences, these two methods have not been compared in the same paper using a common notation. This makes it difficult to understand where the two methods overlap in their assumptions and when one method might be preferred over the other.

To fill this gap, the current thesis provides a unified presentation of these two methods, highlighting their assumptions with a common notation. The performance of the proposed methods was evaluated through a simulation study using the data generating process suggested in Blackwell and Yamauchi (2024). Synthetic datasets

were generated for three different ratios of individuals to time points under two different confounding strengths. The estimates obtained from both methods were compared to the true causal effects to assess their bias, variance and confidence interval coverage.

Beyond the simulation study, a detailed derivation of the loss function for SeqGPLVM is provided, which was not fully elaborated in the original paper. Moreover, a modular implementation of SeqGPLVM is developed using the `GPyTorch` library (Gardner, Pleiss, Bindel, Weinberger, & Wilson, 2021). This implementation is designed to accommodate different treatment types and intended for practical reuse in future research¹.

The remainder of the thesis is structured as follows. Chapter 2 reviews the relevant literature on longitudinal causal inference and the methods exist to address confounding. Next in Chapter 3, the FE-MSM and the SeqGPLVM are described in detail, including their assumptions and estimation procedures. Chapter 4, presents a simulation study comparing the performance of these two methods under different scenarios. Finally, chapter 5, discusses the implications of findings and suggest directions for future research.

¹Code available at https://github.com/Ali-Setareh/SeqGPLVM_FEMSM

2 Background and related work

Methods for observational longitudinal causal inference with discrete time points can be organized along two main dimensions: the structure of the causal process and the causal target of interest. The structure of the causal process is typically represented by a directed acyclic graph (DAG) that encodes the assumptions about the data generating process and the causal relationships between the variables (Pearl, 2009). The presence or absence of treatment-confounder feedback and unobserved confounders are important features of the causal structure that can affect the choice of the estimation method. The causal target of interest, on the other hand, is driven by the research question and scientific context of the study (Lundberg, Johnson, & Stewart, 2021). Different methods may be designed to estimate different causal targets, such as the average treatment effect at a specific time point, the cumulative effect of treatment over time, or the average contrast between two specific treatment regimes. Having these two dimensions in mind, in the following sections, we will review the relevant literature and concepts on longitudinal causal inference and the methods exist to address confounding.

2.1 Conceptual foundations of longitudinal causal inference.

Potential outcomes and treatment regimes The causal targets are often formalized in the potential-outcomes framework (Rubin, 1974). In this framework, the potential outcome of individual i is defined as $Y_i(a)$, representing the outcome that would be observed if the treatment A were set to a specific value a . Note the difference between the observed outcome Y_i and the potential outcome $Y_i(a)$ which is an unobserved variable, as we can only observe the outcome under the treatment that was actually assigned to. Similarly, in the longitudinal setting, we can define the potential outcome at time t for a static treatment regime¹ $\bar{\mathbf{a}}_t = (a_1, a_2, \dots, a_t)$ as $Y_{i,t+1}(\bar{\mathbf{a}}_t)$, representing the outcome that would be observed if the treatment were set to specific values at each time point. In single terminal outcome settings with T time points, it is common to drop the time index on the potential outcome and simply write $Y_i(\bar{\mathbf{a}}_T)$ to represent the potential outcome at the end of the study

¹The static here is to contrast with a dynamic treatment regime, where treatments depend on the values of the confounders (Moodie, Richardson, & Stephens, 2007). In this thesis we only consider static treatment regimes.

under the treatment regime $\bar{\mathbf{a}}_T$. Similarly, the potential value of confounders at time t can also be defined as $X_{it}(\bar{\mathbf{a}}_{t-1})$, representing the value of the confounder at time t that would be observed if the treatment were set to specific values at each time point up to $t - 1$ ².

Treatment-confounder feedback. Treatment-confounder feedback occurs when the treatment at time t can affect the confounders at later time points. This creates a feedback loop between the treatment and the confounders, which can lead to bias if not properly adjusted for; (see Hernán and Robins (2023) for detailed examples.) In Figure 2.2 (b), these feedback loops are represented by the purple arrows from A_t to subsequent $\mathbf{X}_{t+k}$ where $1 \leq k \leq T - t$. These structures are problematic for standard regression-based methods which estimate models such as $\mathbb{E}[Y_i | \bar{\mathbf{A}}_{i3}, \bar{\mathbf{X}}_{i3}] = \alpha + \sum_{t=1}^3 \beta_t A_{it} + \sum_{t=1}^3 \gamma_t^\top \mathbf{X}_{it}$, even if this outcome regression is correctly specified. To see why, in Figure 2.2 (b), $\mathbf{X}_3$ is a confounder for the effect of A_3 on Y as it is a common cause of both A_3 and Y . So it needs to be adjusted for to get an unbiased estimate of the effect of A_3 on Y . However, since $\mathbf{X}_3$ is on the path of A_2 to Y , by conditioning on $\mathbf{X}_3$ part of the causal effect of A_2 on Y mediated through $\mathbf{X}_3$ gets blocked. So methods are needed to both adjust for the time-varying confounders while not blocking the causal paths from the earlier treatments.

Causal target. In longitudinal settings, different causal targets can be defined. One of the most common causal targets is the average of the final potential outcome under a specific treatment regime, i.e., $\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)]$, as a fundamental causal target. It represents the expected final outcome if all individuals in the population were to follow the treatment regime $\bar{\mathbf{a}}_T$. This causal estimand is of particular interest as it can be used to construct various causal contrasts between different treatment regimes. For example, the average causal effect of taking a binary treatment at all-time points compared to not taking the treatment at all can be defined as $\mathbb{E}[Y_i(\bar{\mathbf{1}}_T) - Y_i(\bar{\mathbf{0}}_T)]$.

G-methods. There are well-established methods to identify and estimate $\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)]$, when treatment-confounder feedback is present. The g-formula and marginal structural models (MSMs) are two members of the Robins' g-methods family that aim

²Here we follow this indexing convention to emphasize the temporal ordering of the treatments, confounders and (possibly repeated) outcome. Confounders at time t are measured after the treatment at time $t - 1$ and before the treatment at time t . Moreover, the potential outcome is indexed at time $t + 1$ to reflect the fact that the outcome is determined after the treatment while the measured outcome at time t may act as a confounder for the treatment at time t and the outcomes at later time points.

directly at this causal target (Robins et al., 2000; Shinozaki & Suzuki, 2020; Snowden, Rose, & Mortimer, 2011). While the g-formula gives the full map $\bar{\mathbf{a}} \mapsto \mathbb{E}[Y_i(\bar{\mathbf{a}})]$ for all possible treatment regimes, MSMs are an imposed (restricted) model on $\mathbb{E}[Y_i(\bar{\mathbf{a}})]$. This is particularly useful when the number of time points in a study is large since the number of possible treatment regimes grows exponentially. To see this, if we have a binary treatment at each time point, then there are 2^T possible treatment regimes. This means for a study with 10 time points, there are already 1024 possible treatment regimes. The problem becomes even more severe when the treatment is not binary.

Identification under sequential ignorability. G-methods are relying on the assumptions of sequential ignorability for identification, which states that there are no unmeasured confounders between the treatment and outcome at each time point. More formally, sequential ignorability can be expressed as

$$Y_i(\bar{\mathbf{a}}_T) \perp A_{it} | \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it} \quad \text{for all } t \in \{1, \dots, T\}. \quad (2.1)$$

This equation states that the potential outcome is independent of the treatment at time t given the past treatment and confounder history. This assumption is strong and often violated in observational settings. For instance, in electronic health record data, the socio-economic status of patients is often an important confounder that affects both the treatment decisions and the health outcomes, but it is often not measured in the data. For this reason, the standard methods may not be valid in many real-world applications, which motivates the need for methods that can address unmeasured confounding in longitudinal data.

2.2 Marginal structural models

MSMs are models for average potential outcomes under different treatment regimes. Although these models can be nonparametric (Neugebauer & van der Laan, 2007), they are often specified as a parametric model for $\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)]$:

$$\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)] = m(\bar{\mathbf{a}}_T; \boldsymbol{\theta}_0)$$

where we use $\boldsymbol{\theta}_0$ to denote the set of true parameters. These models are called marginal structural models because they model the mean of the marginal distribution of the potential outcomes, rather than the usual conditional mean of the observed outcomes given the treatments and confounders, i.e., $\mathbb{E}[Y_i | \bar{\mathbf{A}}_T, \bar{\mathbf{X}}_T]$, used

in association studies. A common example of a MSM is the linear model

$$\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)] = m(\bar{\mathbf{a}}_T; \boldsymbol{\theta}) = \theta_1 + \theta_2 \sum_{t=1}^T a_t.$$

Unlike the pure association models, the parameters of MSMs have a causal interpretation. If this model is correctly specified, θ_1 is the expected outcome when no treatment is given at any time point, and θ_2 represents the average causal effect of receiving one more treatment on the expected outcome.

MSMs parameters are often estimated using inverse probability of treatment weighting (IPTW) (Robins et al., 2000). IPTW creates a pseudo-population in which the treatment assignment is exogenous while preserving the causal effect of $\bar{\mathbf{A}}$ on Y . In this pseudo-population, the parameters of the MSM can be estimated using standard regression methods. The way IPTW works is by weighting each observation by the inverse of the probability of receiving the treatment, given their past treatment and confounder history. For binary treatments, let $\pi_{it} = \mathbb{P}(A_{it} = 1 | \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it})$ be the true probability of receiving the treatment at time t given the past treatment and confounder history for individual i (the propensity score). Then the IPTW weight for individual i is defined as $W_i = \prod_{t=1}^T \pi_{it}^{-A_{it}} (1 - \pi_{it})^{-(1-A_{it})}$. Using these weights, Robins (1999) showed that

$$\mathbb{E}[\mathbf{1}\{\bar{\mathbf{A}}_{i,T} = \bar{\mathbf{a}}_T\} W_i Y_i] = m(\bar{\mathbf{a}}_T; \boldsymbol{\theta}_0). \quad (2.2)$$

In observational studies, the true probabilities π_{it} are unknown and need to be estimated from the data. This is typically done by fitting a parametric model for the treatment assignment mechanism. Let $\pi_{it}(\bar{\mathbf{a}}_{t-1}, \bar{\mathbf{x}}_t; \boldsymbol{\beta}_0) = \mathbb{P}(A_{it} = 1 | \bar{\mathbf{A}}_{i,t-1} = \bar{\mathbf{a}}_{t-1}, \bar{\mathbf{X}}_{it} = \bar{\mathbf{x}}_t)$ be a parametrization of the treatment assignment mechanism, with $\boldsymbol{\beta}_0$ denoting the true parameter. The estimated propensity scores are then given by $\hat{\pi}_{it} = \pi_{it}(\bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}; \hat{\boldsymbol{\beta}})$ where $\hat{\boldsymbol{\beta}}$ is the maximum likelihood estimate of $\boldsymbol{\beta}_0$. The estimated IPTW weights are

$$\widehat{W}_i = \prod_{t=1}^T \hat{\pi}_{it}^{-A_{it}} (1 - \hat{\pi}_{it})^{-(1-A_{it})}. \quad (2.3)$$

Once the weights are estimated, the IPTW estimator of MSM can be obtained by solving empirical version of the following equation:

$$\mathbb{E}\left\{\widehat{W}_i \mathbf{h}(\bar{\mathbf{A}}_{i,T}) \left(Y_i - m(\bar{\mathbf{A}}_{i,T}; \boldsymbol{\theta})\right)\right\} = 0, \quad (2.4)$$

where $\mathbf{h}(\cdot)$ is a researcher-chosen vector valued function of the treatment history with dimensionality of $1 \times \dim(\boldsymbol{\theta})$. If $\mathbf{h}(\cdot)$ is chosen to be the identity function and the MSM is specified as a linear model, then the above estimating equation

reduces to a weighted least squares problem.

The IPTW weights can become very large when the estimated propensity scores be close to zero or one, Which may lead to unstable estimates. To mitigate this issue, stabilized weights are often used. They are defined as

$$\widehat{SW}_i = \prod_{t=1}^T \frac{\hat{\pi}_{it}^{A_{it}} (1 - \hat{\pi}_{it})^{1-A_{it}}}{\hat{\pi}_{it}^{A_{it}} (1 - \hat{\pi}_{it})^{1-A_{it}}}. \quad (2.5)$$

In the numerator, $\hat{\pi}_{it}$ is the estimated reduced propensity score, that is, the probability of receiving the treatment at time t given only the past treatment history,

$$\bar{\pi}_{it} = \mathbb{P}(A_{it} = 1 | \bar{\mathbf{A}}_{i,t-1}).$$

The consistency of the IPTW estimator only relies on the correct specification of the treatment assignment model appearing in the denominator of the stabilized weights. The numerator serves only to improve the finite sample performance of the estimator, but the misspecification of it does not affect the consistency. Moreover, any function of the past treatment history can be included in the numerator as long as it is not a function of the confounders (Robins, 2000).

2.3 Time-invariant unmeasured confounding in longitudinal studies

2.3.1 Fixed-effects approaches

Using fixed-effect regression to adjust for unmeasured time-invariant confounders is a common approach in longitudinal data analysis (Imai & Kim, 2019). The idea is to include an individual-specific intercept in the outcome model and estimate the model using the within-individual variation over time. The simplest version of this approach when there is no time-varying confounder is to fit a linear model of the form

$$Y_{i,t+1} = \alpha_i + \beta A_{it} + \epsilon_{it},$$

with α_i being the individual-specific intercept that captures the effect of any unmeasured time-invariant confounder in an additive form.

Although these models can adjust for unmeasured time-invariant confounders, They have several shortcomings. They are primarily designed for settings with repeated measured outcomes, and are not directly applicable to designs with a single terminal outcome. Depending on the context, this can be a significant limitation, as many longitudinal studies are interested in the effect of treatments

on a final outcome measured at the end of the study. More importantly even when there are repeated measures of the outcome, the causal structures in which fixed-effect regression can be used to adjust for unmeasured confounding to get valid estimates of causal effects are limited. Specifically, they can only be used when there is no treatment-confounder feedback. Another important drawback of fixed-effect regression is their inability to target the average potential outcome under specific treatment regimes, which is often of interest in longitudinal studies. Instead, they target the average spontaneous effect of the current treatment on the current outcome, i.e., $\mathbb{E}[Y_{i,t+1}(a_{it} = 1) - Y_{i,t+1}(a_{it} = 0)]$. The assumptions required implicitly by the fixed-effect regression to be causally interpretable is discussed in detail by Imai and Kim (2019).

In contrast, MSMs can target the average potential outcome under a full treatment regime while allowing for complex causal paths. However, they cannot adjust for unmeasured confounding. The FE-MSM proposed by Blackwell and Yamauchi (2024) can be best understood as a hybrid of these approaches. The novelty of FE-MSM comes from its use of a fixed-effect in the propensity score model as opposed to the outcome model. This allows the method to adjust for unmeasured time-invariant confounders without relying on repeated outcome measures, while still benefiting from the ability of MSMs to adjust for all sorts of causal dynamics including treatment-confounder feedback. However, this comes at the cost of targeting the average potential outcome under truncated treatment interventions instead of the full treatment regimes.

2.3.2 Deconfounder theory and sequential deconfounding

Multi-treatment deconfounder

Another approach to address time-invariant unmeasured confounding is based on the deconfounder theory proposed by (Wang & Blei, 2019a). This method is originally devised for static settings with multiple treatments. The key idea is that if there is a true structural unobserved confounder vector $\mathbf{U}_i$ that affects multiple treatments, then it leaves a statistical signature in the observed treatments by inducing conditional independence between the treatments:

$$\mathbb{P}(A_{i1}, A_{i2}, \dots, A_{iM} \mid \mathbf{U}_i, \mathbf{X}_i) = \prod_{j=1}^M \mathbb{P}(A_{ij} \mid \mathbf{U}_i, \mathbf{X}_i)$$

Moreover, if $\mathbf{U}_i$ was observed, then the strong ignorability assumption would hold:

$$Y_i(\bar{\mathbf{a}}_M) \perp (A_{i1}, A_{i2}, \dots, A_{iM}) \mid \mathbf{U}_i, \mathbf{X}_i$$

where M is the number of treatments, A_{ij} is the j -th treatment for individual i , and $\mathbf{X}_i$ is the observed covariate vector.

The deconfounder recipe is then to fit a latent factor model of the assigned treatments which renders the same treatment conditional independence structure as the true unobserved confounder $\mathbf{U}_i$ by substituting the latent variable $\mathbf{Z}_i$ instead:

$$\mathbb{P}_{\Theta}(A_{i1}, A_{i2}, \dots, A_{iM} \mid \mathbf{Z}_i, \mathbf{X}_i) = \prod_{j=1}^M \mathbb{P}(A_{ij} \mid \mathbf{Z}_i, \mathbf{X}_i, \gamma_j)$$

where γ_j is the treatment-specific parameter. Under the assumptions below and SUTVA, this latent variable $\mathbf{Z}_i$ can be used as a substitute for the unobserved confounder $\mathbf{U}_i$ and to retain the ignorability assumption:

$$Y_i(\bar{\mathbf{a}}_M) \perp (A_{i1}, A_{i2}, \dots, A_{iM}) \mid \mathbf{Z}_i, \mathbf{X}_i.$$

Note that the substitute confounder $\mathbf{Z}_i$ is not necessarily the same as the true unobserved confounder $\mathbf{U}_i$.

- **Assumption 1: No single-treatment confounder.** There is no unmeasured confounder that affects only one treatment. Or in other words, all unmeasured confounders are multi-treatment confounders that affect at least two treatments. This assumption is also known as single-cause ignorability assumption.
- **Assumption 2: Consistency of the substitute confounders.** The latent variable $\mathbf{Z}_i$ is pinpointed almost surely by a single deterministic function of the observed treatments:

$$\mathbb{P}(\mathbf{Z}_i \mid A_{i1}, A_{i2}, \dots, A_{iM}) = \delta_{\mathbf{f}(A_{i1}, A_{i2}, \dots, A_{iM})}.$$

Where $\delta_{\mathbf{f}(\cdot)}$ denotes a point mass at $\mathbf{f}(\cdot)$. This means that the latent variable $\mathbf{Z}_i$ is a deterministic function of the observed treatments, and it implies the uniqueness of the latent variable (Wang & Blei, 2020).

- **Assumption 3: Positivity or overlap.** The last k treatments satisfy overlap, $\mathbb{P}(\mathbf{A}_{i,(M-k+1):M} = \mathbf{a}_{i,(M-k+1):M} \mid \mathbf{Z}_i, \mathbf{X}_i) > 0$ for all $\mathbf{a}_{i,(M-k+1):M}$ in the support of $\mathbf{A}_{i,(M-k+1):M}$ and for some $k \geq 1$. This assumption is needed for identification of the average truncated potential outcomes, $\mathbb{E}[Y_i(\mathbf{A}_{1:M-k}, \mathbf{a}_{i,(M-k+1):M})]$, where only the last k treatments are intervened on while the remaining treatments are kept at their observed values.

Debates on multi-treatment deconfounder

Since the introduction of the multi-treatment deconfounder, it has initiated several extensive debates in the literature about the validity of the assumptions. Most of the criticisms are about the consistency assumption and its tension with the positivity assumption. D’Amour (2019) notes that factorization existence does not guarantee the uniqueness of the latent variable and derives an equivalence class of latent variables that can all render the same treatment conditional independence structure, while resulting in different causal effects ranging from negative to positive. This counterexample shows that consistency assumption should be understood as a property of the underlying data generating process and not as consequence of a particular factor model. Hence, it is not possible to be verified by the observed data.

Another point raised by D’Amour (2019) is that, if the consistency assumption holds, there is no variability in the treatments given the latent variable which violates the positivity assumption. Hence, consistency and positivity assumptions cannot hold simultaneously. However, if the causal target of interest is the average potential outcome under interventions on subsets of the treatments, then the positivity assumption can still hold for that subset while the consistency assumption holds for the full set of treatments as stated by Theorem 6 in Wang and Blei (2019a).

The other criticism is concerning whether the deconfounder’s assumptions are truly weaker than the standard ignorability assumption. Grimmer, Knox, and Stewart (2023) argues that in most settings where the deconfounder assumptions hold, the naive regression adjustment with only the observed covariates also results in unbiased estimates of the causal effects. Moreover, through extensive simulation studies, they showed that both the deconfounder method and the naive regression adjustment perform equally poorly in presence of unmeasured confounding.

Furthermore, Ogburn, Shpitser, and Tchetgen (2019) notes that if $\mathbf{Z}_i$ is a deterministic function of the observed treatments, i.e., $\mathbf{Z}_i = \mathbf{f}(A_{i1}, A_{i2}, \dots, A_{iM})$, then $\mathbf{Z}_i$ is a post-treatment variable and conditioning on it may cause biased estimates of the causal effects.

Some of these criticisms are responded by Wang and Blei (2019b) and Wang and Blei (2020). However, the ambiguity surrounding the consistency assumption and the constraints it imposes on the substitute confounder, still remains a major concern for the deconfounder method to be widely applicable.

Longitudinal extensions of the deconfounder

The deconfounder method has been extended to longitudinal settings by Bica, Alaa, and Van Der Schaar (2020) and Hatt and Feuerriegel (2024). They both rely on the same key idea as the original deconfounder method, which is to use a latent variable to capture the statistical signature of the unobserved confounders in the observed treatments. Their main difference lie in the causal structures they are designed for and the causal targets they can identify. While Bica et al. (2020) is designed for settings with multiple treatments at each time point to adjust for time-varying unmeasured confounders using dynamic latent variables, Hatt and Feuerriegel (2024) is designed for a common setting with a single treatment at each time point in presence of constant unmeasured confounding, and it estimates a constant latent variable using the treatment history. Moreover, they both required the consistency assumption which is the most controversial assumption and the implementations of it will be discussed in Chapter 5. A graphical comparison of the sequential deconfounder and the multi-treatment deconfounder is presented In Figure 2.1.

2.4 Research gap and thesis contribution

The literature reviewed in this chapter shows that longitudinal causal inference methods differ not only in the causal structures they are designed to handle, but also in the causal targets they aim to identify. Standard g-methods such as marginal structural models provide a principled way to estimate average potential outcomes under treatment regimes in the presence of treatment-confounder feedback. However, their validity depends on sequential ignorability, an assumption that may be implausible in many observational studies when important baseline characteristics are unmeasured. Fixed-effect approaches and deconfounder-based methods offer two different strategies for addressing time-invariant unmeasured confounding, but they do so under different assumptions and with different implications for the causal estimands that can be identified.

The present thesis is positioned at this intersection. It compares two recent approaches for estimating causal effects of time-varying treatments in the presence of time-invariant unmeasured confounding: the fixed-effect marginal structural model and the sequential Gaussian process latent variable model. Rather than treating these methods as unrelated contributions from different research traditions, this thesis presents them within a common notation and highlights their similarities and differences in terms of their assumptions, estimation procedures, and causal targets. In doing so, it clarifies how the fixed-effect and latent-variable perspectives each

attempt to recover information about unobserved heterogeneity from longitudinal treatment histories. These contributions aim to make the relationship between the two methods more transparent and to support future methodological work on longitudinal causal inference with time-invariant unmeasured confounding.

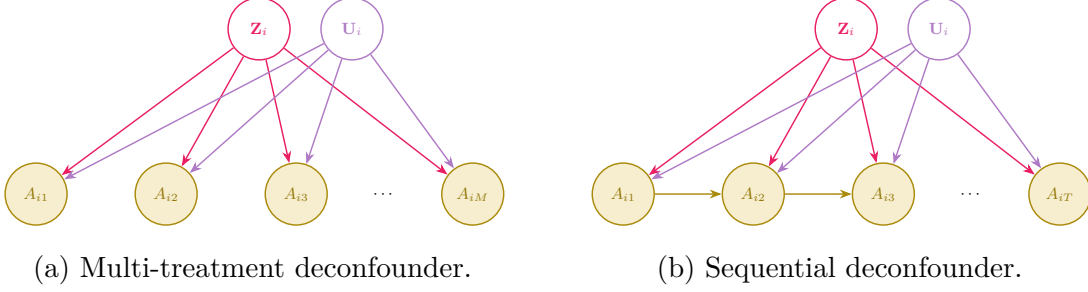
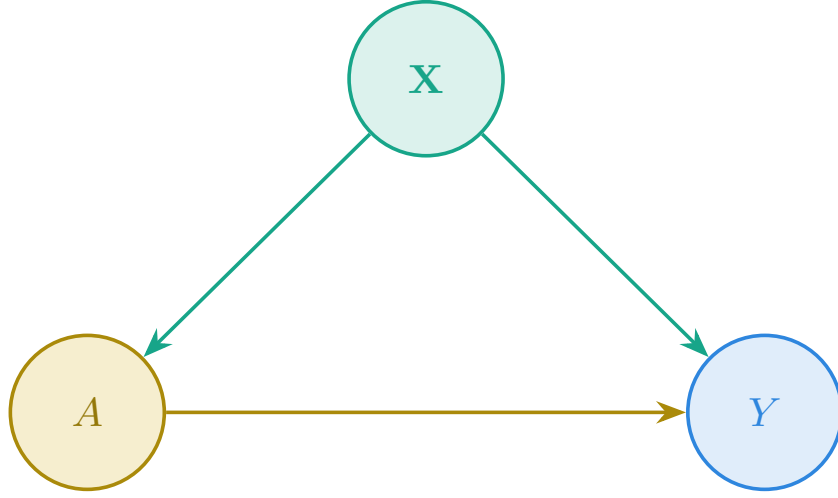
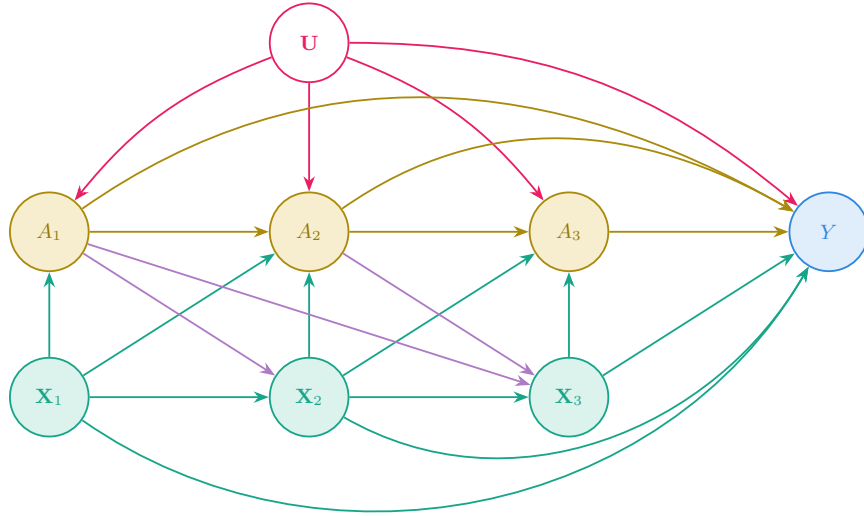


Figure 2.1: A graphical comparison of deconfounder and sequential deconfounder. The first DAG (a) represents the deconfounder setting where there are M causes at a single time point. The second DAG (b) represents the sequential deconfounder setting where there are time-varying T treatments temporally linked. Both DAGs include an unobserved *multi-treatment* confounder $\mathbf{U}_i$ that affects at least two treatments, and a latent variable $\mathbf{Z}_i$ which is a statistically sufficient substitute for the unobserved confounder $\mathbf{U}_i$. Both figures can be extended to include observed confounders.



(a) A simple causal DAG with treatment A , outcome Y , and treatment-outcome confounder $\mathbf{X}$. Confounder $\mathbf{X}$ is a common cause of both the treatment A and the outcome Y , which can lead to confounding bias if not properly adjusted for.



(b) A longitudinal causal DAG with time-varying treatments A_t , time-varying confounders $\mathbf{X}_t$, and an unobserved confounder $\mathbf{U}$. Filled nodes represent observed variables, while the unfilled node represents an unobserved variable. Arrows from A_t to $\mathbf{X}_{t+1}$ represent treatment-confounder feedback, where the treatment at time t can affect the confounders at later time points.

Figure 2.2: Two examples of causal DAGs. The first DAG (a) represents a simple static setting with a single treatment and outcome, while the second DAG (b) represents a more complex longitudinal setting with time-varying treatments.

3 Methods

In this section, we describe the two proposed methods for adjusting for unmeasured time-invariant confounding: the fixed-effect marginal structural model (FE-MSM) proposed by Blackwell and Yamauchi (2024) and the sequential Gaussian process latent variable model (SeqGPLVM) proposed by Hatt and Feuerriegel (2024) when it is used in conjunction with IPTW estimation of MSM (SeqGPLVM-MSM). We first describe the common set of assumptions required by of these methods, and then go into the details of the steps of each method.

3.1 Notation and problem setup

We use uppercase letters for random quantities and the corresponding lowercase letters for their realizations. Boldface denotes vectors or collections, and plain letters denote scalars. Individuals are indexed by $i \in \{1, \dots, N\}$ and time by $t \in \{1, \dots, T\}$. Let $\mathbf{X}_{it} \in \mathbb{R}^D$ denote the observed covariate vector for individual i at time t , and let $A_{it} \in \mathcal{A}_t$ denote the treatment at time t . In this thesis, treatments are assumed to be binary, so that $A_{it} \in \{0, 1\}$, although most notation and derivations extend directly to more general treatment spaces. We adopt the convention that treatment is absent at the beginning of the study, so $A_{i0} = 0$ for all i . The terminal outcome is denoted by Y_i .

For each time point t , define the collections $\mathbf{X}_t := \{\mathbf{X}_{it}\}_{i=1}^N$ and $\mathbf{A}_t := \{A_{it}\}_{i=1}^N$. The full covariate and treatment collections are $\mathbf{X} := \{\mathbf{X}_t\}_{t=1}^T$ and $\mathbf{A} := \{\mathbf{A}_t\}_{t=1}^T$, with realizations $\mathbf{x}$ and $\mathbf{a}$, respectively. The treatment and covariate histories up to time t are written as $\bar{\mathbf{A}}_{it} := (A_{i1}, \dots, A_{it})$ and $\bar{\mathbf{X}}_{it} := (\mathbf{X}_{i1}, \dots, \mathbf{X}_{it})$. For a treatment regime $\bar{\mathbf{a}}_T = (a_1, \dots, a_T)$, the corresponding potential outcome is denoted by $Y_i(\bar{\mathbf{a}}_T)$.

We use $\mathbf{U}_i \in \mathbb{R}^C$ to denote the true structural unmeasured confounder for individual i , assumed to be time-invariant with dimension C . In the latent-variable formulation, we write $\mathbf{Z} := \{\mathbf{Z}_i\}_{i=1}^N$ for the collection of time-invariant latent variables, where $\mathbf{Z}_i \in \mathbb{R}^J$ and J is the latent dimension. The truncated treatment and covariate histories over the intervention window from $T - k$ through T are defined as

$$\underline{\mathbf{A}}_{i,T-k} := (A_{i,T-k}, \dots, A_{iT}), \quad \underline{\mathbf{X}}_{i,T-k} := (\mathbf{X}_{i,T-k}, \dots, \mathbf{X}_{iT}).$$

For $\underline{\mathbf{a}}_{T-k} = (a_{T-k}, \dots, a_T)$, the corresponding truncated potential outcome is defined by

$$Y_i(\underline{\mathbf{a}}_{T-k}) \equiv Y_i(\bar{\mathbf{A}}_{i,T-k-1}, \underline{\mathbf{a}}_{T-k}),$$

that is, the treatment values in the final $k + 1$ -point intervention window are intervened on while the earlier treatment history is left as observed.

Finally, in the context of SeqGPLVM, γ denotes generative hyperparameters and ψ variational hyperparameters. Generative hyperparameters are those that define how the data is generated and enter the prior and likelihood, e.g., the kernel length scale and variance in GPs, while variational hyperparameters are those that define the variational family, e.g., the mean and variance in Gaussian variational distributions.

Target causal estimand. To avoid the usual incidental parameters problem in the estimation of non-linear fixed effects models, FE-MSM is limited to estimating only the average truncated potential outcome. Although, the sequential deconfounding approach and SeqGPLVM can be used for identification and estimation of a wider range of causal targets including $\mathbb{E}[Y_i(\bar{\mathbf{a}}_T)]$, to maintain comparability with FE-MSM, we also focus on the average truncated potential outcome, $\mathbb{E}[Y_i(\underline{\mathbf{a}}_{T-k})]$, as the causal target of interest in this thesis.

3.2 Common assumptions

The common assumptions for both FE-MSM and SeqGPLVM-MSM are stated below. The assumptions can be categorized into two groups: identification assumptions and modeling assumptions. Identification assumptions, are the assumptions which are needed to write target causal estimands such as $\mathbb{E}[Y_i(\underline{\mathbf{a}}_{T-k})]$ in terms of observed data. Based on the identification strategy and the estimator of choice, certain quantities must be modeled. For example, estimating a marginal structural model for the target causal estimand using inverse probability of treatment weighting (IPTW) requires modeling the propensity scores. Modeling assumptions are about the requirements placed on these models in order to have unbiased or consistent estimates. Throughout the text, we use C, I, and M to denote common, identification, and modeling assumptions, respectively.

Assumption C-I1. (*Consistency*). The observed outcome for each individual under their observed treatment history equals the potential outcome under that treatment history. Formally, if $\bar{\mathbf{A}}_{iT} = \bar{\mathbf{a}}_{iT}$, then $Y_i = Y_i(\bar{\mathbf{a}}_{iT})$. Moreover, $Y_i = Y_i(\underline{\mathbf{a}}_{i,T-k})$ if $\bar{\mathbf{A}}_{iT} = (\bar{\mathbf{A}}_{i,T-k-1}, \underline{\mathbf{a}}_{i,T-k})$.

Assumption C-I2. (*No interference*). The treatment assignment for one individual does not affect the potential outcomes of other individuals. This assumption is also known as the no spillover assumption.

Assumption C-I3. (*Time-invariance of unmeasured confounding*). The unmeasured confounding is assumed to be time-invariant. That is, $\mathbf{U}_{it} = \mathbf{U}_i$ for $t \in \{1, \dots, T\}$. We can then write conditional ignorability as

$$\{Y_{i,t+1}(\bar{\mathbf{a}}_t), \bar{\mathbf{X}}_{i,t+1}(\bar{\mathbf{a}}_t)\} \perp\!\!\!\perp A_{it} \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1} = \bar{\mathbf{a}}_{t-1}, \mathbf{U}_i.$$

Note that this assumption is about the true underlying data-generating process and cannot be used for identification directly since $\mathbf{U}_i$ is unmeasured.

Assumption C-I4. (*Positivity*). For all treatment times $t = T - k, \dots, T$, and all histories $(\bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}, \boldsymbol{\xi}_i)$ with positive probability, every treatment level has positive probability:

$$\mathbb{P}(A_{it} = a_t \mid \bar{\mathbf{X}}_{it} = \bar{\mathbf{x}}_t, \bar{\mathbf{A}}_{i,t-1} = \bar{\mathbf{a}}_{t-1}, \boldsymbol{\xi}_i) > 0 \quad \text{for all } a_t \in \mathcal{A}_t$$

Where $\boldsymbol{\xi}_i$ represents the unit-specific fixed effect in FE-MSM or the latent variable in the SeqGPLVM model. This assumption particularly is necessary for the identification of $\mathbb{E}[Y_i(\mathbf{a}_{T-k})]$.

3.3 FE-MSM

FE-MSM consists of two stages: (i) a treatment model that estimates the propensity scores, conditioning on both observed covariates and individual fixed effects that capture the unmeasured time-invariant confounding, and (ii) a marginal structural model that uses the estimated propensity scores in stage one to adjust for confounding using IPTW. FE-MSM can be seen as a very direct way of bringing back the sequential ignorability assumption where we suspect of unmeasured time-invariant confounding. The fixed effect acts as a unit specific offset in the propensity score model, capturing all the time-invariant unmeasured confounding. More formally, the identification of the causal effect of interest relies on the following assumption:

Assumption FE-I1. (*Unit-specific ignorability*). Let α_i be a time-invariant, unidimensional, unmeasured random variable. For all i , t , and $\bar{\mathbf{a}}_T$,

$$\{Y_i(\bar{\mathbf{a}}_T), \bar{\mathbf{X}}_{i,t+1}(\bar{\mathbf{a}}_t)\} \perp\!\!\!\perp A_{it} \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1} = \bar{\mathbf{a}}_{t-1}, \alpha_i.$$

Moreover, the time-invariant

Assumption FE-I1 states that conditioned on the past treatment and covariate

history, as well as the unmeasured fixed effect α_i , the treatment assignment at time t is independent of the future potential outcomes and covariates. Under this assumption along with the assumptions C-I1 to C-I4, the truncated potential outcome $\mathbb{E}[Y_i(\mathbf{a}_{T-k})]$ can be nonparametrically identified as follows

$$\mathbb{E}[Y_i(\mathbf{a}_{T-k})] = \mathbb{E} \left[\frac{\mathbf{1}(\mathbf{A}_{i,T-k} = \mathbf{a}_{T-k}) Y_i}{\prod_{t=T-k}^T \pi_{it}^{a_t} (1 - \pi_{it})^{1-a_t}} \right]. \quad (3.1)$$

Where $\pi_{it} = \mathbb{P}(A_{it} = 1 \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}, \alpha_i)$. In practice, a parametric model such as $\pi_{it}(\bar{\mathbf{x}}_{it}, \bar{\mathbf{a}}_{i,t-1}; \boldsymbol{\beta}_0, \alpha_{i0}) = \mathbb{P}(A_{it} = 1 \mid \bar{\mathbf{X}}_{it} = \bar{\mathbf{x}}_{it}, \bar{\mathbf{A}}_{i,t-1} = \bar{\mathbf{a}}_{i,t-1}; \boldsymbol{\beta}_0, \alpha_{i0})$, is specified; where $\boldsymbol{\beta}_0$ and α_{i0} are the true $b \times 1$ parameter vector and the unit-specific fixed effect respectively. This model is estimated via a conditional maximum likelihood estimator. The log-likelihood function for the treatment model can be written as

$$\ell_{it}(\boldsymbol{\beta}, \alpha_i) = A_{it} \log \pi_{it}(\boldsymbol{\beta}, \alpha_i) + (1 - A_{it}) \log \{1 - \pi_{it}(\boldsymbol{\beta}, \alpha_i)\}.$$

Let $\boldsymbol{\alpha}_0 = (\alpha_1, \dots, \alpha_N)$, then $\boldsymbol{\beta}_0$ and $\boldsymbol{\alpha}_0$ are the solutions to the following optimization problem:

$$(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) = \arg \max_{\boldsymbol{\beta}, \alpha_1, \dots, \alpha_N} \sum_{i=1}^N \sum_{t=1}^T \mathbb{E}[\ell_{it}(\boldsymbol{\beta}, \alpha_i) \mid \alpha_i].$$

The estimates $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \dots, \hat{\alpha}_N)$ and $\hat{\boldsymbol{\beta}}$ are obtained by solving the finite-sample analog of this optimization problem.

Having obtained the estimates of the propensity scores, the weights for the IPTW estimation of the marginal structural model are constructed as

$$\widehat{W}_i(\hat{\boldsymbol{\beta}}, \hat{\alpha}_i) = \prod_{t=T-k}^T \left(\frac{1}{\pi_{it}(\hat{\boldsymbol{\beta}}, \hat{\alpha}_i)} \right)^{A_{it}} \left(\frac{1}{1 - \pi_{it}(\hat{\boldsymbol{\beta}}, \hat{\alpha}_i)} \right)^{1-A_{it}} \quad (3.2)$$

where only the final intervention window from $T - k$ through T is included in the product. However, the propensity score model is estimated using the full treatment history.

If a marginal structural model is specified as $\mathbb{E}[Y_i(\mathbf{a}_{T-k})] = m(\mathbf{a}_{T-k}; \boldsymbol{\theta}_0)$, let

$$\Psi_N(\boldsymbol{\theta}) := \frac{1}{N} \sum_{i=1}^N \widehat{W}_i(\hat{\boldsymbol{\beta}}, \hat{\alpha}_i) \mathbf{h}(\mathbf{A}_{i,T-k}) [Y_i - m(\mathbf{A}_{i,T-k}; \boldsymbol{\theta})]. \quad (3.3)$$

Then, the estimator $\hat{\boldsymbol{\theta}}$ is defined as a solution to the sample estimating equation

$$\Psi_N(\boldsymbol{\theta}) = 0.$$

In Eq. (3.3), $\mathbf{h}(\cdot)$ is a user-specified function of the treatment history. If the marginal structural model is specified as a linear model, a common choice is to set $h(\underline{\mathbf{A}}_{i,T-k}) = \underline{\mathbf{A}}_{i,T-k}$. Then the parameters of the marginal structural model can be estimated via a weighted least squares regression of Y_i on $\underline{\mathbf{A}}_{i,T-k}$ using the weights defined in Eq. (3.2).

The consistency of the $\hat{\boldsymbol{\theta}}$ relies on the correct specification of the propensity score model up to the unmeasured fixed effect α_i .

Assumption FE-M1. (*Correct specification of the parametric component of the propensity score model*). Suppose there exists a finite-dimensional parameter vector $\boldsymbol{\beta}_0$ and unit-specific effects α_{i0} such that, for all i and t ,

$$\mathbb{P}(A_{it} = 1 \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}, \alpha_i) = \pi_{it}(\bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}; \boldsymbol{\beta}_0, \alpha_{i0}) \quad \text{a.s.}$$

That is, conditional on the unit-specific effect α_i , the treatment assignment mechanism is correctly specified as a member of the posited parametric family in $(\bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1})$.

This assumption only requires the propensity score to be correctly specified up to the unit-specific effect α_i , and it does not impose any assumptions on the relationship between α_i and the observed covariates. Based on this assumption and other regularity assumptions, Blackwell and Yamauchi (2024) showed in Theorem 1 that

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(0, V_{\boldsymbol{\theta}_0}),$$

where $V_{\boldsymbol{\theta}_0}$ is a sandwich-type asymptotic variance that can be consistently estimated using the sample analog of the sandwich formula. Figure 3.1, shows a graphical representation of the conditional treatment assignment likelihood.

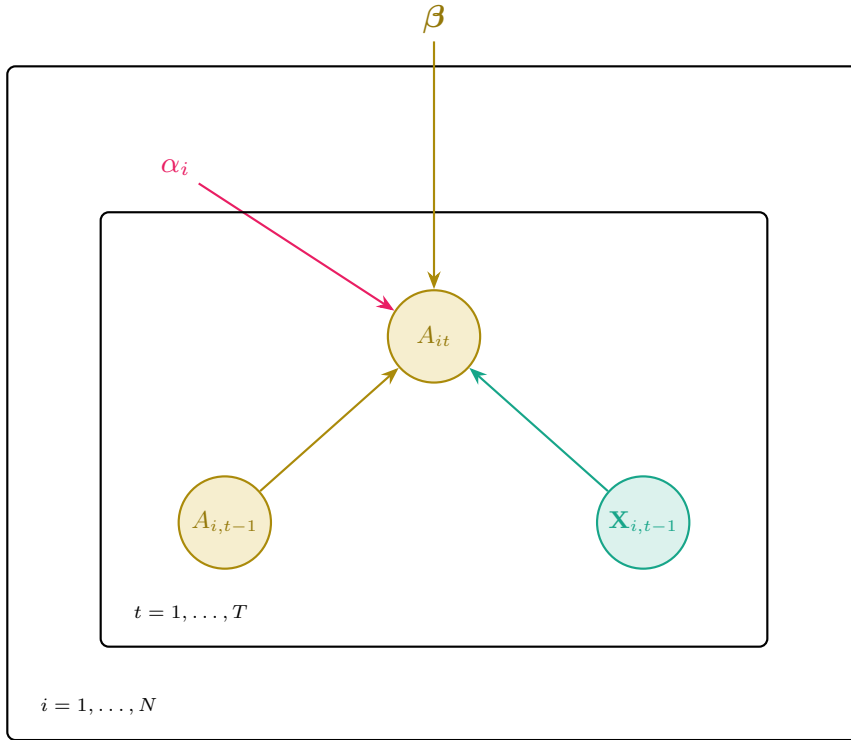


Figure 3.1: Plate diagram of the conditional assignment likelihood under the fixed-effects propensity-score model, assuming the true treatment assignment mechanism depends only on past treatment and covariate values. The likelihood factorizes as $p(\mathbf{a} \mid \mathbf{x}; \alpha, \beta) = \prod_{i,t} p(A_{it} \mid A_{i,t-1}, X_{i,t-1}, \alpha_i, \beta)$. We do not specify a model for the joint distribution of $(\mathbf{X}, \alpha)$. Note that, since α_i is considered as a parameter and not a random variable, it does not have a node around it.

3.4 SeqGPLVM-MSM

SeqGPLVM in combination with MSM consists of three stages: (i) a latent variable model (SeqGPLVM) that estimates a substitute for the unmeasured confounder based on the observed treatment sequence, (ii) estimating the propensity scores using a treatment model that conditions on both observed covariates and the confounder substitute, and (iii) a marginal structural model that uses the estimated propensity scores in stage two to adjust for confounding using IPTW. It is worth mentioning that SeqGPLVM can be used in conjunction with any other estimation methods as well, such as g-computation or doubly robust methods, but we focus on IPTW estimation of MSM for comparability with FE-MSM.

3.4.1 Sequential deconfounding

The sequential deconfounding theory proposed by Hatt and Feuerriegel (2024) can be seen as the sequential version of the multi-treatment deconfounder proposed by Wang and Blei (2019a). Similarly to the multi-treatment deconfounder, it is based on the three key assumptions that are described below.

Assumption SG-I1. (*CMM(p): Conditional Markov Model of order p*). A latent variable $\mathbf{Z}_i \in \mathbb{R}^J$, a fixed order $p \in \mathbb{N}^*$ and a parameter vector $\gamma = (\gamma_1, \dots, \gamma_T)$ exist such that the assigned treatment distribution can be factorized as:

$$p_\gamma(\bar{\mathbf{a}}_{iT} \mid \bar{\mathbf{x}}_{iT}, \mathbf{z}_i) = \prod_{t=1}^T p_{\gamma_t}(a_{it} \mid \mathbf{a}_{i,t-p:t-1}, \bar{\mathbf{x}}_{\leq t}, \mathbf{z}_i). \quad (3.4)$$

for $t < 1$, $\mathbf{a}_t$ is set to 0 by convention. We then say that the latent variable $\mathbf{Z}_i$ follows a Conditional Markov Model of order p , written $\mathbf{Z}_i \sim \text{CMM}(p)$.

In Eq. (3.4), $\gamma = \{\gamma_1, \dots, \gamma_T\}$ are the generative parameters of the treatment assignment model. Note that the conditional probability distribution at each time point can have different parameters. This is crucial to understand the SeqGPLVM architecture.

Assumption SG-I2. (*No unmeasured single-time-point confounding*). All confounders that only affect at a single time point are observed. Equivalently, any unobserved confounder influencing A_t must also induce dependence with at least one other treatment A_s ($s \neq t$), conditional on the observed history.

This assumption excludes structures such as the one in Figure 3.2. Note that, Assumption C-I3 is not sufficient to exclude these single-time-point confounders since a confounder can be constant, but its effect varies with time.

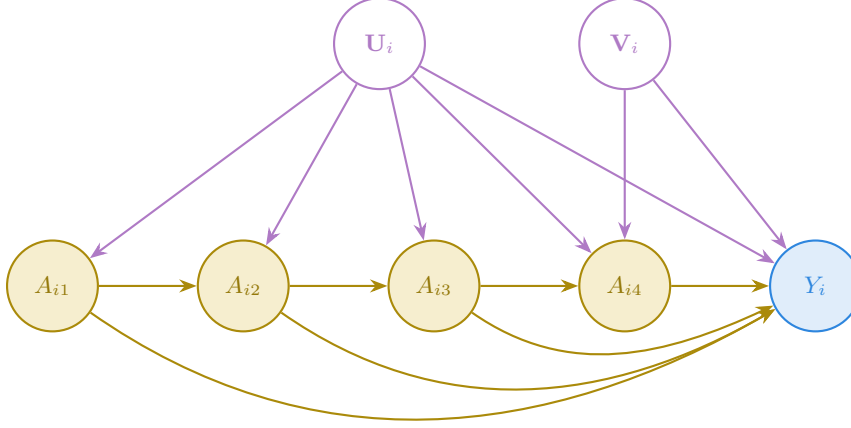


Figure 3.2: Sequential deconfounder excludes unmeasured single-time-point confounders such as V_i .

Assumption SG-I3. (*Consistency of the substitute confounder*). The sequential deconfounder assumes that all latent variables $\mathbf{Z}_i \sim \text{CMM}(p)$ for a particular p are pinpointed by a single deterministic function of treatment and observed covariates history. Formally, there exists a measurable function f such that

$$p(\mathbf{z}_i \mid \bar{\mathbf{x}}_{iT}, \bar{\mathbf{a}}_{iT}) = \delta_{f(\bar{\mathbf{x}}_{iT}, \bar{\mathbf{a}}_{iT})}.$$

Assumption SG-I3 implies the uniqueness of the latent variable $\mathbf{Z}_i$ which imposes the CMM of order p . This assumption is necessary since only the existence of a $\text{CMM}(p)$, does not guarantee the uniqueness of the latent variable. The violation of this assumption can lead to contradictory causal estimates (D’Amour, 2019). However, the substitute confounder $\hat{\mathbf{Z}}_i$ is not required to coincide with the true confounder $\mathbf{U}_i$ or the true latent factorizer $\mathbf{Z}_i$. It is sufficient to estimate the substitute confounder up to some deterministic bijective transformation, e.g., a linear transformation.

Theorem 1 (Substitute Confounder (Hatt & Feuerriegel, 2024)¹). *Under Assumptions SG-I2 and SG-I3, if the treatment process can be represented as a finite-order p CMM, then it follows that sequential ignorability holds conditional on the observed covariate history, treatment history, and latent variables $\mathbf{Z}_i$. Hence, for all $t \in \{1, \dots, T\}$ and all $\bar{\mathbf{a}}_{it}$,*

$$Y_i(\bar{\mathbf{a}}_T) \perp A_{it} \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}, \mathbf{Z}_i. \quad (3.5)$$

¹The original version of this theorem in Hatt and Feuerriegel (2024) is stated only for a first-order CMM. However, we show in Appendix A that the proof of their Theorem 1 using Proposition 7.6 in Kallenberg (1997), can be easily extended to higher-order CMMs as well.

This theorem makes a bridge between the observational world and the potential outcomes world. More importantly, this theorem does not restrict the class of models that can be used to estimate the substitute confounder. It holds for any latent variable model that can estimate a latent variable $\mathbf{Z}_i$ in a CMM of chosen order p . The key idea of the deconfounder is that once the substitute confounder is estimated, we can treat it as a *observed-like* covariates and include it in any outcome model—depending on the specific causal target—to adjust for confounding.

3.4.2 SeqGPLVM

Sequential Gaussian process latent variable model (SeqGPLVM) is a specific implementation of the sequential deconfounding theory based on the Gaussian process latent variable model (GPLVM) (Lawrence, 2003; Titsias & Lawrence, 2010). GPLVM is a non-parametric probabilistic latent variable model that can be used to estimate a low-dimensional latent representation of high-dimensional data. SeqGPLVM extends GPLVM to the sequential setting by modeling the treatment assignment mechanism as a CMM using Gaussian processes.

The objective of SeqGPLVM is to estimate the posterior distribution of the latent variable $\mathbf{Z}_i$ with a chosen dimension J , given the observed treatment history and covariates, $p(\mathbf{z}_i \mid \bar{\mathbf{a}}_{iT}, \bar{\mathbf{x}}_{iT})$, while imposing the CMM structure on the treatment assignment mechanism. To estimate this posterior using the Bayesian framework, a prior and a likelihood model are specified. We assign the following prior to the latent variables assuming they are i.i.d. across individuals and the latent dimensions:

$$p(\mathbf{z}) = \prod_{i=1}^N p(\mathbf{z}_i), \quad \text{where} \quad \mathbf{Z}_i \sim \mathcal{N}(0, I_J) \quad \text{for } i = 1, \dots, N.$$

The SeqGPLVM defines a generative mapping, f_t , from the joint space of the latent confounder substitutes, observed previous treatment history and covariates, to the treatment assignment at each time point:

$$f_t : \mathcal{Z} \times \mathcal{X}_1 \times \dots \times \mathcal{X}_{\leq t} \times \mathcal{A}_{t-p} \times \dots \times \mathcal{A}_{t-1} \rightarrow \mathcal{A}_t.$$

We model these mappings as non-parametric functions using Gaussian processes (GPs), i.e., $f_t \sim \mathcal{GP}(0, k_t(\cdot, \cdot))$ for $t = 1, \dots, T$, where k_t is the kernel function at time t . Since the kernels are defined on the joint space of the latent substitutes and the observed history, there is a lot of flexibility to encode different types of interactions between these variables. For the sake of simplicity, we use the same kernel for all-time points, but in principle, we can use different kernels at different time points. The generative model then can be written as follows:

$$p(\bar{\mathbf{a}}_{iT}, \bar{\mathbf{f}}_T \mid \bar{\mathbf{x}}_{iT}, \mathbf{z}_i) = \prod_{t=1}^T p(a_{it} \mid f_t) p(f_t \mid \mathbf{z}_i, \bar{\mathbf{x}}_{i,\leq t}, \bar{\mathbf{a}}_{i,t-p:t-1}).$$

Since the exact posterior distribution is intractable, we adopt the variational inference with inducing point for inference (Titsias & Lawrence, 2010). The dimensionality of the inducing points is $J + \sum_{s=1}^t |\mathcal{X}_s| + \sum_{s=t-p}^{t-1} |\mathcal{A}_s|$. The likelihood term $p(a_{it} \mid f_t)$ depends on the type of treatment variable. For instance if $\mathcal{A}_t = \{0, 1\}$, we can use a Bernoulli likelihood, or for a continuous treatment, we can use a Gaussian likelihood. The details of the variational inference procedure and the optimization of the evidence lower bound (ELBO) are provided in Appendix B. In Fig 3.3 a graphical representation of the SeqGPLVM is shown.

3.4.3 Outcome model

Once the approximate posterior distribution of the substitute confounder is estimated, it can be used to adjust for confounding in the outcome model. This stage closely parallels the FE-MSM procedure described in Section 3.3. In particular, Theorem 1 implies identification of the mean truncated potential outcome over the last k time periods through the same IPTW representation as in Eq. (3.1), with the unobserved fixed effect α_i replaced by the latent variable $\mathbf{Z}_i$:

$$\mathbb{E}[Y_i(\mathbf{a}_{T-k})] = \mathbb{E} \left[\frac{\mathbf{1}(\mathbf{A}_{i,T-k} = \mathbf{a}_{T-k}) Y_i}{\prod_{t=T-k}^T \pi_{it}^{a_t} (1 - \pi_{it})^{1-a_t}} \right], \quad (3.6)$$

where

$$\pi_{it} = \mathbb{P}(A_{it} = 1 \mid \bar{\mathbf{X}}_{it}, \bar{\mathbf{A}}_{i,t-1}, \mathbf{Z}_i).$$

Hence, as in FE-MSM, estimation proceeds in two steps: first, a parametric model is used to estimate the propensity scores, and second, these estimated propensities are used to construct inverse-probability weights for fitting the marginal structural model. The key difference is that instead of directly estimating a unit-specific fixed effect α_i in the treatment model, a sample is drawn from the approximate posterior distribution of the substitute confounder $q(\mathbf{Z}_i)$ and condition the treatment model on these samples together with the observed history to estimate the propensity scores.

To account for uncertainty in the inferred substitute confounder, we adopt a multiple-imputation-style procedure. Specifically, for each posterior draw $\mathbf{z}_i^{(s)} \sim q(\mathbf{Z}_i)$, $s = 1, \dots, S$, the propensity scores are re-estimated and the corresponding

IPTW weights are computed as,

$$\widehat{W}_i(\hat{\beta}, \mathbf{z}_i^{(s)}) = \prod_{t=T-k}^T \left(\frac{1}{\pi_{it}(\hat{\beta}, \mathbf{z}_i^{(s)})} \right)^{A_{it}} \left(\frac{1}{1 - \pi_{it}(\hat{\beta}, \mathbf{z}_i^{(s)})} \right)^{1-A_{it}},$$

and fit the same marginal structural model as in the FE-MSM approach. Let

$$\Psi_N(\boldsymbol{\theta}_s) := \frac{1}{N} \sum_{i=1}^N \widehat{W}_i(\hat{\beta}, \mathbf{z}_i^{(s)}) \mathbf{h}(\mathbf{A}_{i,T-k}) \left[Y_i - m(\mathbf{A}_{i,T-k}; \boldsymbol{\theta}_s) \right].$$

For each draw s , the corresponding MSM parameter estimate $\hat{\boldsymbol{\theta}}^{(s)}$ is obtained by solving

$$\Psi_N(\boldsymbol{\theta}_s) = 0.$$

Equivalently, when the MSM is linear and $\mathbf{h}(\mathbf{A}_{i,T-k})$ is chosen appropriately, this step can be implemented as a weighted least squares regression using the draw-specific IPTW weights.

The final point estimate is then obtained by averaging across the S posterior draws,

$$\hat{\boldsymbol{\theta}} = \frac{1}{S} \sum_{s=1}^S \hat{\boldsymbol{\theta}}^{(s)}.$$

This procedure introduces two sources of uncertainty: first, uncertainty about the latent substitute confounder induced by the approximate posterior $q(\mathbf{Z}_i)$, and second, the usual sampling uncertainty in the MSM fit conditional on a given latent draw. To propagate both sources into the final causal effect estimates, we combine the draw-specific estimates using Rubin's within-between decomposition (Rubin, 2009). Let $V_j^{(s)}$ denote the robust (sandwich) variance estimate associated with the j -th component of the MSM parameter vector $\hat{\boldsymbol{\theta}}^{(s)}$ for draw s , and let Q_j represent the average of these variances across draws for this component, then the total variance estimator for the j -th component of $\hat{\boldsymbol{\theta}}$ is given by

$$Q_j = \frac{1}{S} \sum_{s=1}^S V_j^{(s)}, \quad B_j = \frac{1}{S-1} \sum_{s=1}^S (\hat{\theta}_j^{(s)} - \bar{\theta}_j)^2, \quad T_j = Q_j + \left(1 + \frac{1}{S}\right) B_j.$$

The resulting confidence interval T_j therefore reflect both (i) variability in the MSM fit conditional on a latent draw and (ii) additional variability due to uncertainty in the inferred substitute confounder.

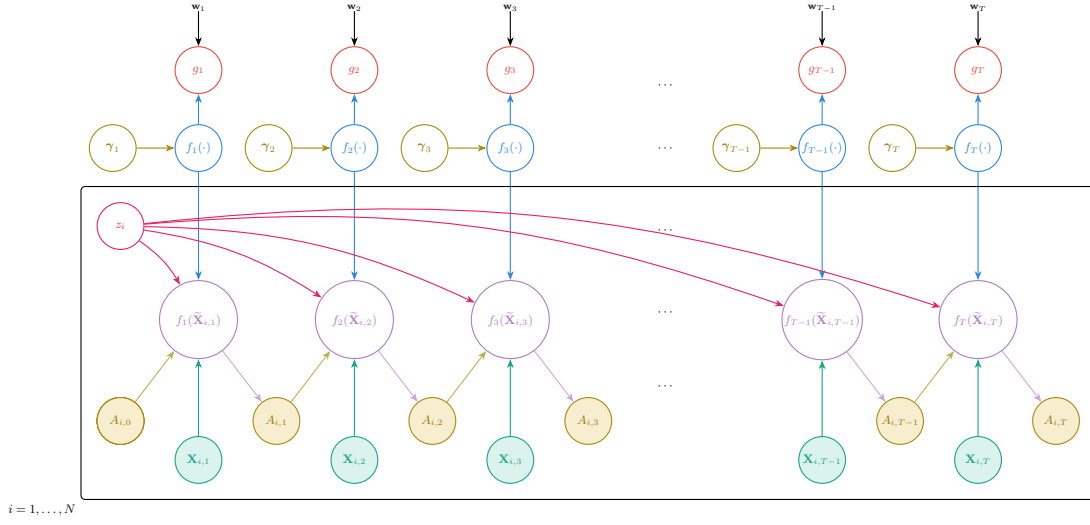


Figure 3.3: Plate diagram of the first-stage SeqGPLVM treatment model used to estimate time-varying propensity scores for a first order CMM. For each time point t , a Gaussian process $f_t(\cdot)$ with hyperparameters γ_t maps the latent state and observed history into the assignment probability A_{it} . The variables g_t denote GP function values at inducing inputs m_t (shown without nodes as they are treated as parameters). Filled nodes indicate observed quantities and unfilled nodes latent quantities. The outer plate indexes individuals $i = 1, \dots, N$ and the temporal structure repeats over $t = 1, \dots, T$. The diagram represents the conditional assignment model underlying propensity-score estimation and does not impose a model on the marginal distribution of the covariate process.

4 Simulation study

4.1 Data generating process

To evaluate the finite sample performance of the two proposed method, we conduct a simulation study based on the DGP taken from Section 5.1 of Blackwell and Yamauchi (2024). In this DGP, the scenario which we simulate is a study design where individuals are exposed to a binary treatment (e.g., receiving a political advertisement) over a period fixed period of time, and the outcome of interest is measured only once at the end of the study period (e.g., voting behavior). The treatment assignment is influenced by both observed covariates and an unmeasured time-invariant confounder (e.g., political leaning). The outcome is affected by the treatment received at the final time point, the cumulative effect of treatments received in the last three time points, the mean of observed covariates over time, and the unmeasured confounder.

We simulate longitudinal data for $N \in \{200, 500, 1000\}$ individuals. For the length of the time series for each N , we consider three different lengths: $\rho = N/T$ with $\rho \in \{5, 10, 50\}$. The data is balanced, meaning that each individual has the same number of time points. In Eq. (4.1), we present the structural causal equations for the DGP. Here, $\text{expit}(x) = \frac{1}{1+e^{-x}}$ and $\bar{\mathbf{X}}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{X}_{it}$.

$$U_i := \varepsilon_i, \quad \varepsilon_i \sim \text{Unif}(-c_U, c_U). \quad (4.1a)$$

$$\mathbf{X}_{it} := -\frac{1}{2}\mathbb{1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \Sigma). \quad (4.1b)$$

$$A_{it} := \mathbb{1}\left\{\eta_{it} \leq \text{expit}\left(U_i + \varphi A_{i,t-1} + \boldsymbol{\beta}^\top \mathbf{X}_{it}\right)\right\}, \quad \eta_{it} \sim \text{Unif}(0, 1) \quad (4.1c)$$

$$Y_i := U_i + \tau_F A_{iT} + \tau_C \sum_{t=T-3}^{T-1} A_{it} + \boldsymbol{\theta}^\top \bar{\mathbf{X}}_i + \epsilon_{iT}, \quad \epsilon_{iT} \sim \mathcal{N}(0, 1) \quad (4.1d)$$

We simulate for two different values of $c_U \in \{1, 2\}$ to control the contribution of the unmeasured confounder U_i . We simulate for two different numbers of covariates $d_X \in \{2, 4\}$ to see the effect of the dimensionality of covariates. The coefficients are set to $\boldsymbol{\beta} = (-0.5, -0.5)^\top$ when $d_X = 2$ and $\boldsymbol{\beta} = (-0.5, -0.5, 1, -0.5)^\top$ when $d_X = 4$. We set $\varphi = 0.3$. The covariates $\mathbf{X}_{it}$ are generated exogenously to the treatments solely for the sake of simplicity where $\Sigma_{jj} = 1$ and $\Sigma_{jj'} = 0.2$ for $j \neq j'$. By doing this, the ground truth values for the causal targets of interests coincide with the DGP parameters. Also note that there is no dynamics in the covariates

and the distribution of $\mathbf{X}_{it}$ is the same for all-time points. The parameters for the outcome model are set to $\tau_F = 1$, $\tau_C = 0.3$, and $\boldsymbol{\theta} = (1, 0.5)^\top$ when $d_X = 2$ and when $d_X = 4$, $\boldsymbol{\theta} = (1, 0.5, 1, 1)^\top$. The causal DAG for the DGP is shown in Figure 4.1.

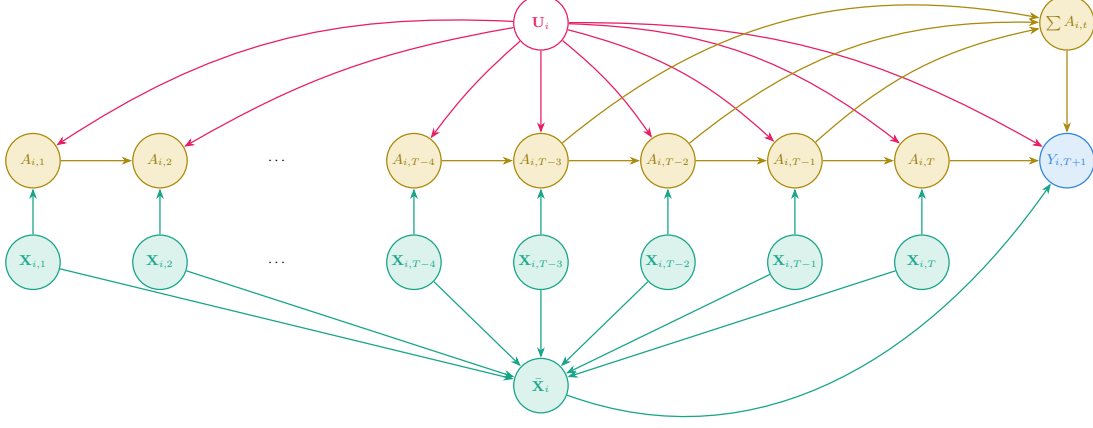


Figure 4.1: Data generating process causal DAG.

4.2 Estimation

For defining our causal targets, first we need to specify the causal questions of interest. Define

$$S_k(\mathbf{a}_{T-k}) := \sum_{t=T-k}^{T-1} a_t \in \{0, 1, 2, \dots, k\}$$

as the sum of treatments received in the last k time points before T , where $k + 1$ represents the intervention window. In real world applications, the choice of k is based on the research question and the domain knowledge about the relevant time window for the treatment effects. In this simulation, we set $k = 3$ to match the dependence structure of the outcome on the treatments in the DGP.

We are interested in two causal questions. What is the average effect of the treatment received at the final time on the final outcome while holding fixed the sum of the earlier treatments within the intervention window, marginalized over the natural values of the treatments this treatment window? Mathematically, this is defined as $\mathbb{E}[Y_i(a_T = 1, S_3 = s) - Y_i(a_T = 0, S_3 = s)]$. The second causal question is: what is the average effect of receiving one additional treatment in the last three time points, before T on the final outcome while keeping the final treatment fixed, marginalized over the natural values of the treatments in this window? To express this mathematically, we can write it as $\mathbb{E}[Y_i(S_3 = s + 1, A_{iT} = a) - Y_i(S_3 = s, A_{iT} = a)]$.

To answer both of these causal questions, a marginal structural model is proposed that relates the assigned treatments to the expected potential outcome at the final time: $\mathbb{E}[Y_i(\mathbf{a}_{T-k})] = m(\mathbf{a}_{T-k}, \boldsymbol{\theta}_0)$, where $m(\mathbf{a}_{T-k}, \boldsymbol{\theta}_0)$ is the true unknown function parameterized by $\boldsymbol{\theta}_0$. Our job is to first propose a function $f(\mathbf{a}_{T-k}, \boldsymbol{\theta})$ and then estimate its parameters $\boldsymbol{\theta}$. In this DGP, the ground truth function m , is:

$$m(\mathbf{a}_{T-k}; \boldsymbol{\theta}_0) = \alpha_0 + \tau_F a_T + \tau_C \sum_{t=T-3}^{T-1} a_t \quad (4.2)$$

To show this, we begin from Eq. (4.1d). First note that since this is a structural equation model, we can replace the random variables by their interventional values. So we have:

$$Y_i(\mathbf{a}_{T-k}) = U_i + \tau_F a_T + \tau_C \sum_{t=T-3}^{T-1} a_t + \boldsymbol{\theta}^\top \bar{\mathbf{X}}_i(\mathbf{a}_{T-k}) + \epsilon_{iT}.$$

Taking the expectation on both sides and making use of the exogeneity of the covariates and treatments, we have:

$$\mathbb{E}[Y_i(\mathbf{a}_{T-k})] = \mathbb{E}[U_i] + \tau_F a_T + \tau_C \sum_{t=T-3}^{T-1} a_t + \boldsymbol{\theta}^\top \mathbb{E}[\bar{\mathbf{X}}_i(\mathbf{a}_{T-k})] + \mathbb{E}[\epsilon_{iT}] \quad (4.3)$$

$$= \mathbb{E}[U_i + \boldsymbol{\theta}^\top \bar{\mathbf{X}}_i(\mathbf{a}_{T-k})] + \tau_F a_T + \tau_C \sum_{t=T-3}^{T-1} a_t \quad (4.4)$$

$$= \alpha_0 + \tau_F a_T + \tau_C \sum_{t=T-3}^{T-1} a_t, \quad \alpha_0 := \mathbb{E}[U_i + \boldsymbol{\theta}^\top \bar{\mathbf{X}}_i(\mathbf{a}_{T-k})] \quad (4.5)$$

From Eq. (4.2), it can be seen:

$$\begin{aligned} \mathbb{E}[Y_i(a_T = 1, S_3 = s) - Y_i(a_T = 0, S_3 = s)] &= \tau_F \\ \mathbb{E}[Y_i(a_T = a, S_3 = s + 1) - Y_i(a_T = a, S_3 = s)] &= \tau_C \end{aligned}$$

The equality between these causal effects and the parameters of the outcome model in the DGP is a result of the linearity of the DGP and the absence of treatment-confounder feedback. To only focus on the adjustment for the unmeasured confounder, for both of the methods, we specify $f(\mathbf{a}_{T-k}; \boldsymbol{\theta})$ as the correctly specified linear MSM:

$$f(\mathbf{a}_{T-k}; \boldsymbol{\theta}) = \alpha' + \tau'_F a_T + \tau'_C \sum_{t=T-3}^{T-1} a_t \quad (4.6)$$

where $\boldsymbol{\theta} = (\alpha', \tau'_F, \tau'_C)$.

4.3 Results

We compared the performance of the proposed methods by estimating the two structural parameters of interest τ_F and τ_C using the weighted least squares. For FE-MSM, the parameters are estimated by solving the following optimization problem:

$$(\hat{\tau}_F, \hat{\tau}_C) = \arg \min_{\tau'_F, \tau'_C} \sum_{i=1}^N \widehat{W}_i \left\{ Y_i - \alpha - \tau'_F A_{iT} - \tau'_C \sum_{t=T-3}^{T-1} A_{it} \right\}^2 \quad (4.7)$$

Where $\widehat{W}_i$ are the stabilized version of IPTW weights defined in Eq. (3.2). The parametric propensity score model is correctly specified (up to a unit-specific fixed effect) as a logistic regression of the treatment on the past treatment history and covariates, with a unit-specific fixed effect,

$$p(a_{it} = 1 \mid \mathbf{x}_{it}, a_{i,t-1}, \alpha_i) = \text{expit}(w_1^\top \mathbf{x}_{it} + w_2 a_{i,t-1} + \alpha_i). \quad (4.8)$$

For the sequential GPLVM, a CMM(1) with a one-dimensional latent variable (i.e., $\mathbf{Z}_i \in \mathbb{R}^J$ with $J = 1$) is fitted to the treatment data, using a linear kernel:

$$k_t^{\text{treat}}((\mathbf{x}_t, a_{t-1}, \mathbf{z}), (\mathbf{x}'_t, a'_{t-1}, \mathbf{z}')) = \nu \left[\sum_{j=1}^{d_X} x_{jt} x'_{jt} + a_{t-1} a'_{t-1} + \sum_{j=1}^J z_j z'_j \right]$$

where ν is the variance hyperparameter. This is equivalent to the Bayesian linear regression with a latent variable. The observed confounders are standardized before fitting the GPLVM. A weakly informative Gamma prior is placed on ν , specifically $\nu \sim \Gamma(2, 1)$. The parameters of the GPLVM are estimated by minimizing the negative evidence lower bound using the Adam optimizer with learning rate set to 0.01. The number of iterations is set to 6000 and the convergence is monitored by checking the change in ELBO. The parameters of the MSM are estimated by the procedure described in Section 3.4.3 by solving the same optimization problem as in Eq. (4.7) with number of posterior draws set to $S = 100$. The following logistic regression model is used to estimate the propensity scores for the IPTW estimation of the MSM parameters:

$$p(a_{it} = 1 \mid \mathbf{x}_{it}, a_{i,t-1}, \mathbf{z}_i^{(s)}) = \text{expit}(w_3^\top \mathbf{x}_{it} + w_4 a_{i,t-1} + w_5^\top \mathbf{z}_i^{(s)} + w_6), \quad (4.9)$$

where $\mathbf{z}_i^{(s)}$ is the s -th posterior draw of the latent variable for individual i .

For both methods, the weighting is stabilized by multiplying the weights with conditional probabilities of the current treatment given only the previous treatment

using the following logistic regression model:

$$p(a_{it} = 1 \mid a_{i,t-1}) = \text{expit}(w_7 a_{i,t-1} + w_8). \quad (4.10)$$

Trimming weights. The unit-specific fixed effects and the individual latent variables are not identified for individuals with no variation in their treatment history, i.e., always treated or never treated. Including these samples in the estimation process can make the optimizations for both methods unstable. Even if the population level positivity assumption holds, in the finite sample, it is possible to have such individuals. This problem becomes more severe when the number of time points is small. In this case, we removed always and never treated individuals from the estimation process and imputed their propensity scores by assigning a fixed value, e.g., $\hat{\pi}_{it} = 0.01$ for never treated individuals and $\hat{\pi}_{it} = 0.99$ for always treated individuals for all-time points. Depending on the length of the treatment intervention window k specified in the MSM, this imputation strategy can lead to extreme weights which we further trimmed by capping them within the range of $[10^{-6}, 1 - 10^{-6}]$ to avoid unstable weights.

Figures 4.2–4.4 show the results of the simulation study for the two covariate-case. The additional results for the four covariate-case are reported in Appendix C. Bias, standard error, and coverage of the 90% confidence intervals are reported based on 500 simulation runs. Figure 4.2 reports the bias, Figure 4.3 reports the standard error, and Figure 4.4 reports the coverage. Within each figure, the first two rows correspond to the weak unmeasured confounding setting where $U_i \in [-1, 1]$ ($c_U = 1$), and the last two rows correspond to the strong unmeasured confounding setting where $U_i \in [-2, 2]$ ($c_U = 2$). The first and third rows correspond to $\tau_F = 1$, while the second and fourth rows correspond to $\tau_C = 0.3$. The columns correspond to the N to T ratio, with panels (A), (D), (G), and (J) showing $\rho = 5$ (the largest number of time points), panels (B), (E), (H), and (K) showing $\rho = 10$, and panels (C), (F), (I), and (L) showing $\rho = 50$ (the smallest number of time points). Solid lines in grey show the results using the true propensity scores (**IPTW-True**). Solid orange lines show the results for FE-MSM (**FE-MSM**) and dash-dotted purple lines show the results for SeqGPLVM (**SeqGPLVM-MSM**). Dashed blue lines represent the results for the naive IPTW estimation of the specified MSM assuming sequential ignorability without adjusting for the unmeasured confounder (**IPTW**). In this case we used the same logistic regression as in Eq. (4.8) to estimate the propensity scores by replacing the individual fixed effects α_i with a global intercept.

The results show that both methods improves the bias of the estimates of the parameters of interest in comparison to the naive IPTW, especially as the ratio of $\frac{N}{T}$ decreases. However, FE-MSM tends to have slightly lower bias and standard

error compared to the SeqGPLVM-MSM, particularly for smaller sample sizes. This may be due to the fact that the FE-MSM method relies on a correctly specified parametric model for the propensity scores, while the sequential GPLVM method is more flexible but may require larger sample sizes to achieve similar performance. We can see that the worst case for both methods is for $\rho = 50$, but specially this is visible for the sequential GPLVM in the smallest sample size $N = 200$, where it introduces more bias than the naive IPTW for the strong unmeasured confounding case (Figure 4.2, panels (I) and (L)) and performs nearly as poorly as the naive IPTW for the weak unmeasured confounding case (Figure 4.2, panels (C) and (F)). However, the larger estimated standard errors in this setting do not generally keep the coverage close to the nominal level. In particular, for the shortest treatment sequence ($\rho = 50$) under the stronger unmeasured confounding setting (Figure 4.4, panels (I) and (L)), the coverage of SeqGPLVM-MSM falls well below 0.90, especially in panel (L), corresponding to the $\tau_C = 0.3$ scenario. Thus, in the worst-case regime, the increase in uncertainty is not sufficient to compensate for the additional bias.

As the number of time points increases, the gap between the bias of two methods decreases, and they perform more similarly although the FE-MSM bias remains lower while the standard error of SeqGPLVM-MSM remains higher than the FE-MSM estimates across all scenarios. Under the weak unmeasured confounding setting where the unmeasured confounder explains roughly 1/3 of the variance of the treatment assignments (Figure 4.4, panels (A)–(F)), coverage is mixed rather than uniformly close to the nominal level. For $\tau_C = 0.3$, FE-MSM attains coverage values below 0.90 throughout, and SeqGPLVM-MSM shows substantial undercoverage for the shortest treatment sequence, with the deterioration being most pronounced for $N = 500$ and $N = 1000$ (panel (F)). For $\tau_F = 1.0$, both methods are generally closer to the nominal level, although some undercoverage remains, again most visibly for $\rho = 50$ (panel (C)). In this weaker confounding setting, FE-MSM is generally closer to IPTW-True than SeqGPLVM-MSM, but it does not match the true-propensity benchmark, especially for the shortest treatment sequence. Even when $c_U = 1$, FE-MSM still exhibits noticeably larger bias and lower coverage than IPTW-True, and this gap becomes more pronounced under stronger unmeasured confounding.

In the stronger unmeasured confoundedness case where the unmeasured confounder explains 2/3 of the variance in the treatment (Figures 4.2–4.4, panels (G)–(L)), both methods perform more poorly than the weaker case in terms of the bias while FE-MSM performs slightly better across all different ρ values and sample sizes. In this scenario there is a small Monte Carlo bias observed even for the IPTW-True, due to extreme weights and the truncation strategy.

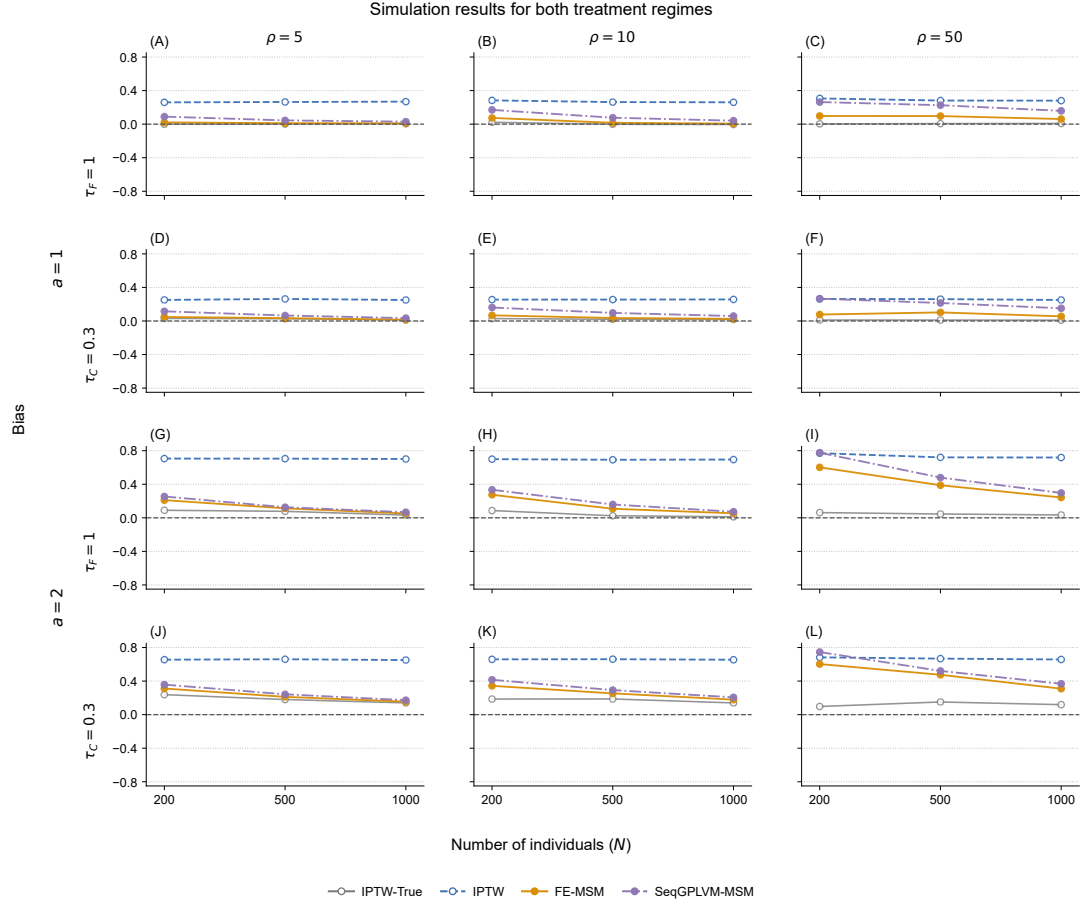


Figure 4.2: Simulation bias for the two-covariate case. The panels report bias for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.

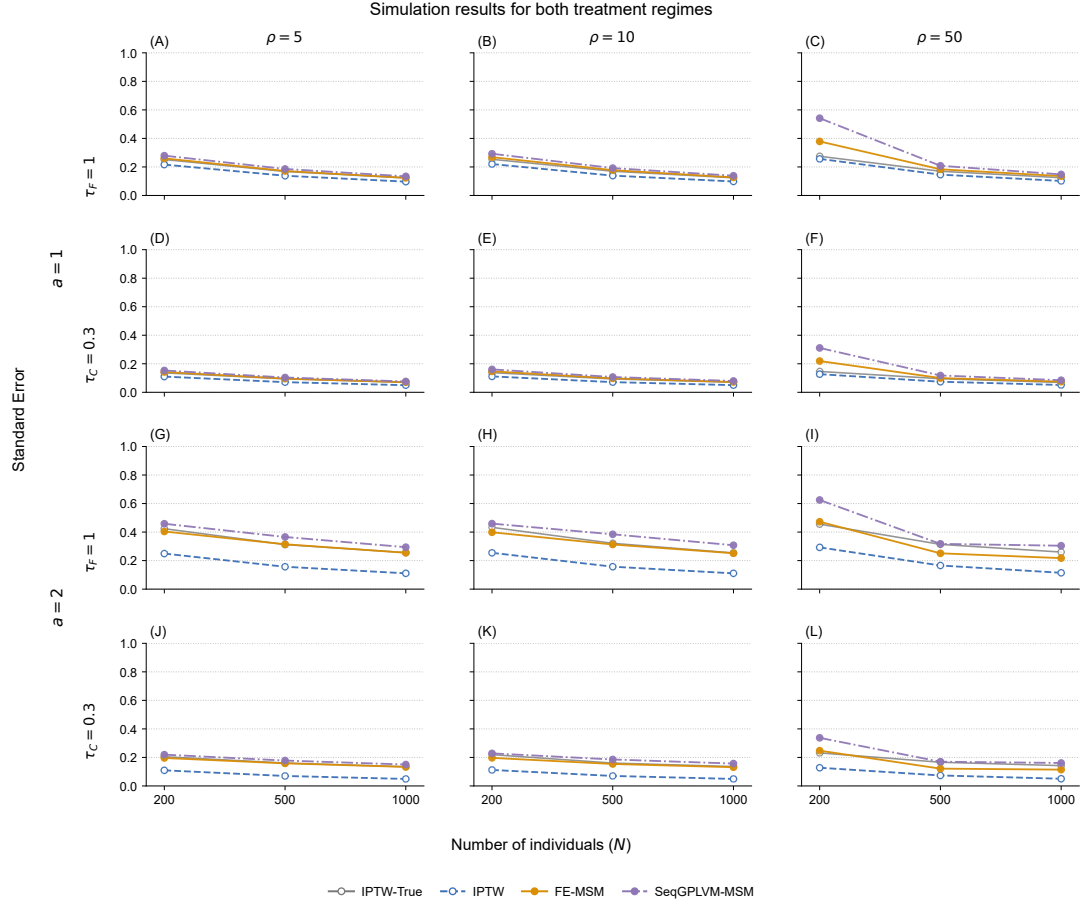


Figure 4.3: Simulation standard error for the two-covariate case. The panels report standard error for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.

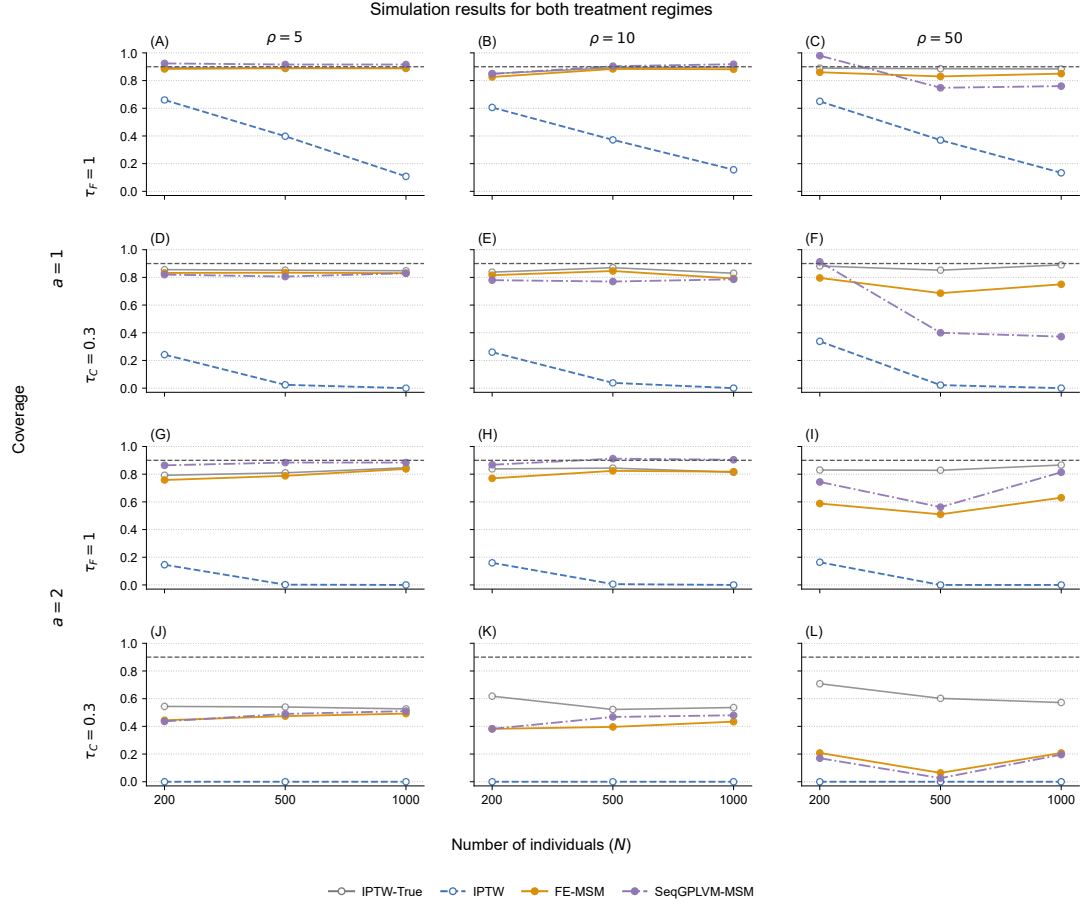


Figure 4.4: Simulation coverage for the two-covariate case. The panels report coverage of the 90% confidence intervals for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.

5 Discussion

This thesis aimed to clarify the assumptions and the finite-sample performance of two approaches to address time-constant unmeasured confounding in the context of time-varying treatments. Although the simulation results show improvements in the bias of the causal effect estimates for both methods, there are still some limitations and open questions which make these methods not yet ready for widespread use in practice. The fundamental requirement behind both of these methods is another form of ignorability assumption that includes a time-invariant latent variable in the conditioning set, making the treatment assignment independent of the potential outcomes.

In FE-MSM this requirement is directly assumed and is formalized by Assumption FE-I1. In particular, the assumption states that all time-invariant unmeasured confounding in the treatment process can be represented by a latent scalar variable, α_i . Effectively, this assumption restricts the class of DGPs which this method can address to the class of DGPs where the unmeasured confounder only enters the treatment process as a scalar or, at least, where those confounders can be summarized into one scalar. This is a strong structural assumption which cannot be verified using observational data. For instance, this method cannot be used in cases where the unmeasured confounding modifies not only the intercept but also the effect of other covariates or treatments (interaction effects). In light of this, the simulation results for FE-MSM should be interpreted cautiously, since the DGP was constructed under conditions that are particularly well suited to this method.

On the other hand, in SeqGPLVM, this requirement is proved by means of assuming a set of auxiliary assumptions (Assumptions SG-I1-SG-I3). Among these assumptions, the assumption of consistency of the latent variable plays the most crucial role. This assumption is not only impossible to verify in real-world problems, but it is not straightforward to establish even in simulation studies where the DGP is known. To verify this assumption for a given DGP, one would need to show that the relevant latent confounder is uniquely identified, up to trivial transformations, by the observed data. Note that the notion of identifiability considered here is closer to the structural identifiability, since the issue is whether the latent confounder is uniquely implied by the data-generating structure, rather than whether a parameter can be estimated within a fixed statistical model. This is a much stronger requirement than merely showing that there exists some latent-variable representation of the treatment process. In

general, different latent structures may induce the same observed distribution of the treatments, and therefore the same observed data may be compatible with multiple candidate substitute confounders. In such a case, even if a latent-variable model fits the treatment process well, it does not follow that the recovered latent variable corresponds to the confounder needed for causal identification. Putting it differently, the fact that SeqGPLVM may recover a latent factor that explains the dependence structure among the treatments, but this alone does not guarantee that this latent factor is the correct one for the causal problem. This is a fundamental issue for all methods that rely on latent variable models for deconfounding, and it is not specific to SeqGPLVM.

Consequently, the favorable simulation results for SeqGPLVM should be interpreted with caution. This method is shown to work in a specific DGP and for larger sample sizes, but the satisfaction of the consistency assumption is left for future work. More broadly, the method shifts the burden of the usual sequential ignorability assumption to a different and arguably less transparent set of structural assumptions about the existence, uniqueness, and causal relevance of a latent substitute confounder.

Despite the theoretical limitations of latent variable models for deconfounding, SeqGPLVM remains a promising approach for predictive tasks and exploratory data analysis. Its flexibility in modeling complex relationships among observed covariates, treatments, and latent confounders in longitudinal settings, together with the ability to incorporate domain knowledge through the choice of kernel, makes it particularly appealing in applications where latent variables are themselves of substantive interest, such as psychometrics. Moreover, the model allows these functional relationships to vary across time points, thereby accommodating greater temporal heterogeneity. A natural direction for future research is to extend this framework to incorporate dynamic latent variables and to implement the amortized version, which may further enhance its usefulness as a nonparametric latent variable model for time-series data.

In conclusion this thesis provided a common framework for understanding the assumptions and the finite-sample performance of two methods for addressing time-constant unmeasured confounding in time-varying treatments settings. It highlighted the assumptions and limitations of these methods and the need for further theoretical and empirical work to understand their properties and applicability. The results of this thesis provided simulation evidence that both FE-MSM and SeqGPLVM can be useful tools for addressing time-constant unmeasured confounding in certain settings to reduce bias and improve coverage compared to standard methods that do not account for unmeasured confounding. A promising future research direction is to investigate the theoretical properties of the fixed

effects estimates by FE-MSM and the latent variables estimated by SeqGPLVM to understand the nature of these estimates and their relationship to the true unmeasured confounder. This could be pursued by further exploring the performance and the estimation of the latent variables and the fixed effects in simulation studies using alternative DGPs that impose a richer structure on the unmeasured confounding. For example, one could consider a hierarchical DGP in which individuals are first assigned to latent classes and the unmeasured confounder is then generated conditional on class membership. Such a design would induce structured heterogeneity in the population and would make it possible to investigate whether the fixed effects in FE-MSM and the latent variables inferred by SeqGPLVM recover only coarse subgroup structure or a quantitatively meaningful approximation to the underlying confounder. In this sense, the simulation framework suggested by Bouchattaoui, Tami, Lepetit, and Cournède (2025) may provide a useful starting point, even though this paper focuses on unobserved adjustment variables rather than unobserved confounders.

Another idea inspired by Bouchattaoui et al. (2025) is to investigate whether the consistency assumption for SeqGPLVM can be imposed also in practice by imposing a more concentrated, approximately deterministic posterior over the latent variables. The theory of deconfounding requires that the latent substitute confounder is a deterministic function of the treatment sequence while in practice the posterior distribution over the latent variables is typically diffuse, especially for smaller sample sizes. If successful, such an approach could lead to more stable latent representations and reduce the sensitivity of the resulting causal effect estimates to posterior uncertainty in the inferred confounder.

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A Proof of Theorem 1

The appendix of Hatt and Feuerriegel (2024) formulates the proof for Theorem 1 only for a first-order conditional Markov model over treatments. However, the same proof strategy extends naturally to finite order p by augmenting the treatment state with the last p treatments. The result below is therefore a straightforward extension of their recursive-construction argument.

Proposition 1 (Finite-order extension of the proof strategy of Hatt and Feuerriegel (2024)). *Suppose that for some fixed integer $p \geq 1$, the assigned treatment process satisfies the conditional Markov property of order p , that is,*

$$p_\gamma(\bar{\mathbf{a}}_{iT} \mid \bar{\mathbf{x}}_{iT}, \mathbf{z}_i) = \prod_{t=1}^T p_{\gamma_t}(a_{it} \mid \mathbf{a}_{i,t-p:t-1}, \bar{\mathbf{x}}_{it}, \mathbf{z}_i), \quad (\text{A.1})$$

where $\mathbf{a}_{i,t-p:t-1} = (a_{i,t-p}, \dots, a_{i,t-1})$, with the convention that for $t < 1$, $a_{it} = 0$. Assume further that the treatment space is a Borel space and that unobserved confounding is time-invariant in the sense maintained by Hatt and Feuerriegel (2024).

Then the proof strategy of Hatt and Feuerriegel (2024) extends to yield a recursive representation

$$A_{it} = f_t(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it}), \quad (\text{A.2})$$

where $V_{it} \sim \text{Uniform}([0, 1])$, and

$$V_{it} \perp Y_i(\bar{\mathbf{a}}_t) \mid \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i \quad \text{for all } \bar{\mathbf{a}}_t \in \bar{\mathcal{A}}_t. \quad (\text{A.3})$$

Consequently,

$$A_{it} \perp Y_i(\bar{\mathbf{a}}_t) \mid \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i \quad \text{for all } \bar{\mathbf{a}}_t \in \bar{\mathcal{A}}_t. \quad (\text{A.4})$$

Proof. The argument follows the same structure as the appendix of Hatt and Feuerriegel (2024), but with a state augmentation step.

First, define the augmented treatment state

$$\mathbf{S}_{it} := (A_{i,t-p+1}, \dots, A_{it}) \in \mathcal{A}^p.$$

Under (A.1), the treatment process is Markov of order p conditional on $(\bar{\mathbf{X}}_{it}, \mathbf{Z}_i)$.

Equivalently, the augmented state process $(\mathbf{S}_{it})_{t \geq 1}$ is first-order Markov conditional on $(\bar{\mathbf{X}}_{it}, \mathbf{Z}_i)$. Indeed, for each t ,

$$\mathbb{P}(\mathbf{S}_{it} \in B \mid \bar{\mathbf{S}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i) = \mathbb{P}(\mathbf{S}_{it} \in B \mid \mathbf{S}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i)$$

for all measurable $B \subseteq \mathcal{A}^p$, because the first $p-1$ coordinates of $\mathbf{S}_{it}$ are deterministic shifts of $\mathbf{S}_{i,t-1}$, while the last coordinate A_{it} depends on the past only through $\mathbf{A}_{i,t-p:t-1}$, $\bar{\mathbf{X}}_{it}$, and $\mathbf{Z}_i$.

Since $\mathcal{A}^p$ is again a Borel space, the same Kallenberg recursion argument used by Hatt and Feuerriegel (2024) for the order-1 case applies to the augmented state process. Therefore, there exist measurable functions F_t and i.i.d. random variables $V_{it} \sim \text{Uniform}([0, 1])$ such that

$$\mathbf{S}_{it} = F_t(\mathbf{S}_{i,t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it}).$$

Let f_t denote the last-coordinate projection of F_t . Then

$$A_{it} = f_t(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it}),$$

which establishes (A.2).

It remains to verify (A.3). This is the same step that remains to be shown in the proof of Lemma 2 in Hatt and Feuerriegel (2024). Their argument extends without essential change. Suppose, for contradiction, that there exists an omitted variable W that confounds V_{it} and $Y_i(\bar{\mathbf{a}}_t)$ conditional on $(\bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i)$. Moreover, according to Assumption SG-I2, there is no single-time confounders. Hence, W at least affect V_{is} where $s \neq t$. But the recursion construction yields that the V_{it} are i.i.d. $\text{Uniform}[0, 1]$, so such a common time-invariant source of dependence is excluded. If instead the omitted variable is time-varying, then because

$$A_{it} = f_t(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it}),$$

the same omitted variable would induce confounding between A_{it} and $Y_i(\bar{\mathbf{a}}_t)$. This contradicts the maintained setup that any unobserved confounding is time-invariant and is captured through the latent static variable $\mathbf{Z}_i$. Hence,

$$V_{it} \perp Y_i(\bar{\mathbf{a}}_t) \mid \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i,$$

which proves (A.3).

Finally, define

$$\mathcal{G}_{it} := \sigma(\bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i).$$

Since $(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it})$ is $\mathcal{G}_{it}$ -measurable, (A.3) implies

$$(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it}) \perp Y_i(\bar{\mathbf{a}}_t) \mid \mathcal{G}_{it}.$$

Because A_{it} is a measurable function of $(\mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i, \bar{\mathbf{X}}_{it}, V_{it})$, conditional independence is preserved under measurable transformations, and therefore

$$A_{it} \perp Y_i(\bar{\mathbf{a}}_t) \mid \mathcal{G}_{it}.$$

Equivalently,

$$A_{it} \perp Y_i(\bar{\mathbf{a}}_t) \mid \bar{\mathbf{A}}_{i,t-1}, \bar{\mathbf{X}}_{it}, \mathbf{Z}_i.$$

This proves (A.4). □

B SeqGPLVM loss function

In this section, we derive the ELBO (evidence lower bound) for the sequential Gaussian process latent variable model (SeqGPLVM) introduced in Section 3.4. We retain the notation from Section 3.1.

B.1 Variational inference background

The ultimate inferential goal of the SeqGPLVM is to estimate the posterior distribution over the time-invariant latent substitute confounder $\mathbf{Z}$ given the observed data $\mathbf{X}$ and $\mathbf{A}$. By Bayes' rule, we can write the posterior as a function of the likelihood and prior:

$$p_{\gamma}(\mathbf{z} \mid \mathbf{x}, \mathbf{a}) = \frac{p_{\gamma}(\mathbf{a} \mid \mathbf{x}, \mathbf{z}) p_{\gamma}(\mathbf{z} \mid \mathbf{x})}{p_{\gamma}(\mathbf{a} \mid \mathbf{x})}.$$

where the denominator $p_{\gamma}(\mathbf{a} \mid \mathbf{x})$ (the *marginal likelihood / evidence*) is the normalizing constant. Note that we use γ to parameterize both the likelihood and the prior to make the notation more concise, but in practice, they can have different hyperparameters. Rewriting the logarithm of evidence as an integral over latent variables,

$$\log p_{\gamma}(\mathbf{a} \mid \mathbf{x}) = \log \int p_{\gamma}(\mathbf{a}, \mathbf{z} \mid \mathbf{x}) d\mathbf{z}. \quad (\text{B.1})$$

In latent-variable models this integral is typically intractable which means the posterior cannot be computed exactly; introducing a tractable variational density $q_{\psi}(\mathbf{z} \mid \mathbf{x})$ yields a lower bound on $\log p_{\gamma}(\mathbf{a} \mid \mathbf{x})$. The gap between this bound and the true log-evidence is a KL divergence to the true posterior, so maximizing the bound pushes $q_{\psi}(\mathbf{z} \mid \mathbf{x})$ towards $p_{\gamma}(\mathbf{z} \mid \mathbf{x}, \mathbf{a})$. To see this, the marginal likelihood can be written as

$$\begin{aligned} \log p_{\gamma}(\mathbf{a} \mid \mathbf{x}) &= \log \int p_{\gamma}(\mathbf{a}, \mathbf{z} \mid \mathbf{x}) d\mathbf{z} \\ &= \log \int q_{\psi}(\mathbf{z} \mid \mathbf{x}) \frac{p_{\gamma}(\mathbf{a}, \mathbf{z} \mid \mathbf{x})}{q_{\psi}(\mathbf{z} \mid \mathbf{x})} d\mathbf{z} \\ &\geq \int q_{\psi}(\mathbf{z} \mid \mathbf{x}) \log \frac{p_{\gamma}(\mathbf{a}, \mathbf{z} \mid \mathbf{x})}{q_{\psi}(\mathbf{z} \mid \mathbf{x})} d\mathbf{z} =: \mathcal{L}(\psi, \gamma), \end{aligned} \quad (\text{B.2})$$

where the inequality is Jensen's inequality. $\mathcal{L}(\psi, \gamma)$ is called the evidence lower bound (ELBO). The KL divergence between the variational distribution and the

true posterior can be written as

$$\begin{aligned}
\text{KL}(q_\psi(\mathbf{z} \mid \mathbf{x}) \parallel p_\gamma(\mathbf{z} \mid \mathbf{x}, \mathbf{a})) &= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log \frac{q_\psi(\mathbf{z} \mid \mathbf{x})}{p_\gamma(\mathbf{z} \mid \mathbf{x}, \mathbf{a})} d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) [\log q_\psi(\mathbf{z} \mid \mathbf{x}) - \log p_\gamma(\mathbf{z} \mid \mathbf{x}, \mathbf{a})] d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \left[\log q_\psi(\mathbf{z} \mid \mathbf{x}) - \log \frac{p_\gamma(\mathbf{a}, \mathbf{z} \mid \mathbf{x})}{p_\gamma(\mathbf{a} \mid \mathbf{x})} \right] d\mathbf{z} \\
&= \log p_\gamma(\mathbf{a} \mid \mathbf{x}) - \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log \frac{p_\gamma(\mathbf{a}, \mathbf{z} \mid \mathbf{x})}{q_\psi(\mathbf{z} \mid \mathbf{x})} d\mathbf{z} \\
&= \log p_\gamma(\mathbf{a} \mid \mathbf{x}) - \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}).
\end{aligned}$$

Equivalently,

$$\log p_\gamma(\mathbf{a} \mid \mathbf{x}) = \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) + \text{KL}(q_\psi(\mathbf{z} \mid \mathbf{x}) \parallel p_\gamma(\mathbf{z} \mid \mathbf{x}, \mathbf{a})). \quad (\text{B.3})$$

Maximizing $\mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma})$ w.r.t. q minimizes the KL divergence to the true posterior and equality holds iff $q_\psi(\mathbf{z} \mid \mathbf{x}) = p_\gamma(\mathbf{z} \mid \mathbf{x}, \mathbf{a})$. This is true since the lhs of (B.3) does not depend on q . Thus, we can use $\mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma})$ as a surrogate objective for approximate inference. $\mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma})$ can be rewritten as

$$\begin{aligned}
\mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) &= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log \left(\frac{p_\gamma(\mathbf{a}, \mathbf{z} \mid \mathbf{x})}{q_\psi(\mathbf{z} \mid \mathbf{x})} \right) d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log p_\gamma(\mathbf{a}, \mathbf{z} \mid \mathbf{x}) d\mathbf{z} - \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log q_\psi(\mathbf{z} \mid \mathbf{x}) d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log \left(p_\gamma(\mathbf{a} \mid \mathbf{x}, \mathbf{z}) p_\gamma(\mathbf{z} \mid \mathbf{x}) \right) d\mathbf{z} - \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log q_\psi(\mathbf{z} \mid \mathbf{x}) d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log p_\gamma(\mathbf{a} \mid \mathbf{x}, \mathbf{z}) d\mathbf{z} - \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log \left(\frac{q_\psi(\mathbf{z} \mid \mathbf{x})}{p_\gamma(\mathbf{z} \mid \mathbf{x})} \right) d\mathbf{z} \\
&= \int q_\psi(\mathbf{z} \mid \mathbf{x}) \log p_\gamma(\mathbf{a} \mid \mathbf{x}, \mathbf{z}) d\mathbf{z} - \text{KL}(q_\psi(\mathbf{z} \mid \mathbf{x}) \parallel p_\gamma(\mathbf{z} \mid \mathbf{x})) \\
&= \mathbb{E}_{q_\psi(\mathbf{z} \mid \mathbf{x})} [\log p_\gamma(\mathbf{a} \mid \mathbf{x}, \mathbf{z})] - \text{KL}(q_\psi(\mathbf{z} \mid \mathbf{x}) \parallel p_\gamma(\mathbf{z} \mid \mathbf{x})). \quad (\text{B.4})
\end{aligned}$$

Thus to setup variational inference, we need to specify the likelihood $p_\gamma(\mathbf{a} \mid \mathbf{x}, \mathbf{z})$, the prior over latents $p_\gamma(\mathbf{z} \mid \mathbf{x})$, and the variational family $q_\psi(\mathbf{z} \mid \mathbf{x})$.

B.2 Conditional markov model

Let $\widetilde{\mathbf{X}}_t = \{(\bar{\mathbf{X}}_{it}, \mathbf{A}_{i,t-p:t-1}, \mathbf{Z}_i)\}_{i=1}^N$ be the augmented covariate vector at time t that includes observed covariates history, the treatment history from time $t - p$ to $t - 1$, and the latent substitute confounder. If the treatment assignments are generated according to a conditional Markov model of order p then the likelihood in Eq. (B.4)

can be factorized as

$$p_{\gamma}(\mathbf{a} \mid \mathbf{x}, \mathbf{z}) = \prod_{t=1}^T p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t). \quad (\text{B.5})$$

Substituting this factorization into the ELBO in Eq. (B.4) yields

$$\mathcal{L}(\psi, \gamma) = \mathbb{E}_{q_{\psi}(\mathbf{z} \mid \mathbf{x})} \left[\sum_{t=1}^T \log p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) \right] - \text{KL}(q_{\psi}(\mathbf{z} \mid \mathbf{x}) \parallel p_{\gamma}(\mathbf{z} \mid \mathbf{x})). \quad (\text{B.6})$$

B.3 Variational Inference for Bayesian Gaussian process latent variable models

Bayesian Gaussian process latent variable models (GPLVMs) (Titsias & Lawrence, 2010) are a class of latent variable models that use Gaussian processes to model the relationship between observed data and latent variables. The term Bayesian is to distinguish it from the original GPLVM (Lawrence, 2003) where the latent variables $\mathbf{Z}$ are point estimated rather than inferred in a fully Bayesian manner. From now on, by GPLVMs we mean Bayesian GPLVMs. The idea behind GPLVMs is to model the relationship between the outcome and the observed covariates and the latent variables as a nonparametric function f with a Gaussian process prior over it,

$$A_{it} \mid \tilde{\mathbf{X}}_{it}, f_t \sim p_t(\cdot \mid \boldsymbol{\lambda}_{it}), \quad l_t(\boldsymbol{\lambda}_{it}) = f_t(\tilde{\mathbf{X}}_{it}).$$

where l_t is an appropriate link function and $\boldsymbol{\lambda}_{it}$ denotes the parameter, or set of parameters, indexing the conditional distribution of A_{it} . The function f_t is a latent function in this framework, since it is not directly observed in the data. Accordingly, it is assigned a prior distribution, namely a Gaussian process,

$$f_t(\cdot) \sim \mathcal{GP}(m_t(\cdot), k_t(\cdot, \cdot; \gamma_t)).$$

where $m_t(\cdot)$ is the mean function, $k_t(\cdot, \cdot; \gamma_t)$ is the kernel function parameterized by hyperparameters γ_t . They are both defined on the augmented input space. Let $\mathbf{f}_t = \{f_t(\tilde{\mathbf{x}}_{it})\}_{i=1}^N$ denote the vector of latent function values at these augmented inputs. The likelihood $p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t)$ is then obtained by marginalizing out the latent

function values,

$$\begin{aligned}
p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) &= \int p_{\gamma_t}(\mathbf{a}_t, \mathbf{f}_t \mid \tilde{\mathbf{x}}_t) d\mathbf{f}_t \\
&= \int p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t, \tilde{\mathbf{x}}_t) p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) d\mathbf{f}_t \\
&= \int p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) d\mathbf{f}_t.
\end{aligned} \tag{B.7}$$

The last equality holds since the treatment is conditionally independent of the covariates and latent variables given the latent function since the covariates only enter the likelihood through the latent function (see Figure 3.3). Here, $p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)$ is the GP prior over the vector of function values at the augmented inputs. If $p(\mathbf{a}_t \mid \mathbf{f}_t)$ is Gaussian, then the marginal likelihood can be computed in closed form:

$$p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) = \mathcal{N}(\mathbf{a}_t \mid m_t(\tilde{\mathbf{x}}_t), k_t(\tilde{\mathbf{x}}_t, \tilde{\mathbf{x}}_t; \gamma_t) + \sigma^2 \mathbf{I}).$$

However, in general, the likelihood is not Gaussian (e.g., binary treatments) and the integral in Eq. (B.7) is intractable. In this case by introducing a variational distribution $q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)$ and applying Jensen's inequality, we can obtain a lower bound on the log-likelihood at each time point:

$$\begin{aligned}
\log p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) &= \log \int p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) d\mathbf{f}_t \\
&= \log \int q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) \frac{p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)}{q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} d\mathbf{f}_t \\
&\geq \int q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) \log \left(\frac{p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)}{q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} \right) d\mathbf{f}_t \\
&= \mathbb{E}_{q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} [\log p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t)] - \text{KL}(q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) \parallel p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)).
\end{aligned}$$

This lower bound is true for all $t \in \{1, \dots, T\}$. Hence, the lower bound on the sum of log-likelihoods across all-time points is

$$\sum_{t=1}^T \log p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) \geq \sum_{t=1}^T \mathbb{E}_{q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} [\log p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t)] - \sum_{t=1}^T \text{KL}(q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) \parallel p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t))$$

Taking the expectation w.r.t. $q_{\psi}(\mathbf{z} \mid \mathbf{x})$ on both sides yields

$$\begin{aligned}
\mathbb{E}_{q_{\psi}(\mathbf{z} \mid \mathbf{x})} \left[\sum_{t=1}^T \log p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) \right] &\geq \sum_{t=1}^T \mathbb{E}_{q_{\psi}(\mathbf{z} \mid \mathbf{x})} \left[\mathbb{E}_{q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} [\log p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t)] \right] \\
&\quad - \sum_{t=1}^T \mathbb{E}_{q_{\psi}(\mathbf{z} \mid \mathbf{x})} \left[\text{KL}(q_{\psi_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) \parallel p(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)) \right].
\end{aligned}$$

Substituting this lower bound into the ELBO in Eq. (B.6) yields

$$\begin{aligned}\mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) &\geq \sum_{t=1}^T \mathbb{E}_{q_{\boldsymbol{\psi}}(\mathbf{z}|\mathbf{x})} \left[\mathbb{E}_{q_{\boldsymbol{\psi}_t}(\mathbf{f}_t|\tilde{\mathbf{x}}_t)} [\log p_{\boldsymbol{\gamma}_t}(\mathbf{a}_t | \mathbf{f}_t)] \right] \\ &\quad - \sum_{t=1}^T \mathbb{E}_{q_{\boldsymbol{\psi}}(\mathbf{z}|\mathbf{x})} \left[\text{KL}\left(q_{\boldsymbol{\psi}_t}(\mathbf{f}_t | \tilde{\mathbf{x}}_t) \parallel p(\mathbf{f}_t | \tilde{\mathbf{x}}_t)\right) \right] \\ &\quad - \text{KL}(q_{\boldsymbol{\psi}}(\mathbf{z} | \mathbf{x}) \parallel p_{\boldsymbol{\gamma}}(\mathbf{z} | \mathbf{x})) = \tilde{\mathcal{L}}(\boldsymbol{\psi}, \boldsymbol{\gamma}).\end{aligned}\tag{B.8}$$

Hence,

$$\log p_{\boldsymbol{\gamma}}(\mathbf{a} | \mathbf{x}) \geq \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) \geq \tilde{\mathcal{L}}(\boldsymbol{\psi}, \boldsymbol{\gamma}).\tag{B.9}$$

Thus, $\tilde{\mathcal{L}}(\boldsymbol{\psi}, \boldsymbol{\gamma})$ is a valid lower bound on the log-evidence and can be used as a surrogate objective for variational inference in the sequential GPLVM.

B.4 Sparse variational approximation via inducing points

The lower bound in Eq. (B.8) still involves a variational distribution over the full random vector of latent function values $\mathbf{f}_t \in \mathbb{R}^N$ at each time point t . In Gaussian process models, this leads to computational costs that scale poorly with the number of observations. To obtain a more scalable approximation, we introduce a set of $M_t \ll N$ inducing inputs for each time point t ,

$$\mathbf{w}_t = \{\mathbf{w}_{mt}\}_{m=1}^{M_t},$$

defined on the same augmented input space as $\tilde{\mathbf{x}}_t$, together with the corresponding inducing variables

$$\mathbf{g}_t = \{f_t(\mathbf{w}_{mt})\}_{m=1}^{M_t}.$$

Under the Gaussian process prior, the joint distribution of the random latent function values $\mathbf{f}_t$ at the observed augmented inputs and the inducing random variables $\mathbf{g}_t$ is Gaussian:

$$\begin{bmatrix} \mathbf{f}_t \\ \mathbf{g}_t \end{bmatrix} \mid \tilde{\mathbf{x}}_t, \mathbf{w}_t \sim \mathcal{N}\left(\begin{bmatrix} \mathbf{m}_{\tilde{\mathbf{x}},t} \\ \mathbf{m}_{\mathbf{w},t} \end{bmatrix}, \begin{bmatrix} \mathbf{K}_{\tilde{\mathbf{x}}\tilde{\mathbf{x}}}^{(t)} & \mathbf{K}_{\tilde{\mathbf{x}}\mathbf{w}}^{(t)} \\ \mathbf{K}_{\mathbf{w}\tilde{\mathbf{x}}}^{(t)} & \mathbf{K}_{\mathbf{w}\mathbf{w}}^{(t)} \end{bmatrix}\right),$$

where

$$\mathbf{m}_{\tilde{\mathbf{x}},t} := m_t(\tilde{\mathbf{x}}_t), \quad \mathbf{m}_{\mathbf{w},t} := m_t(\mathbf{w}_t),$$

and

$$\mathbf{K}_{\tilde{\mathbf{x}}\tilde{\mathbf{x}}}^{(t)} := k_t(\tilde{\mathbf{x}}_t, \tilde{\mathbf{x}}_t; \boldsymbol{\gamma}_t), \quad \mathbf{K}_{\tilde{\mathbf{x}}\mathbf{w}}^{(t)} := k_t(\tilde{\mathbf{x}}_t, \mathbf{w}_t; \boldsymbol{\gamma}_t),$$

$$\mathbf{K}_{\mathbf{w}\tilde{\mathbf{x}}}^{(t)} := k_t(\mathbf{w}_t, \tilde{\mathbf{x}}_t; \gamma_t), \quad \mathbf{K}_{\mathbf{w}\mathbf{w}}^{(t)} := k_t(\mathbf{w}_t, \mathbf{w}_t; \gamma_t).$$

It follows that

$$p_{\gamma_t}(\mathbf{f}_t, \mathbf{g}_t \mid \tilde{\mathbf{x}}_t, \mathbf{w}_t) = p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t),$$

where

$$p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t) = \mathcal{N}(\mathbf{m}_{\mathbf{w},t}, \mathbf{K}_{\mathbf{w}\mathbf{w}}^{(t)}),$$

and

$$\begin{aligned} p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) &= \mathcal{N}\left(\mathbf{m}_{\tilde{\mathbf{x}},t} + \mathbf{K}_{\tilde{\mathbf{x}}\mathbf{w}}^{(t)} (\mathbf{K}_{\mathbf{w}\mathbf{w}}^{(t)})^{-1} (\mathbf{g}_t - \mathbf{m}_{\mathbf{w},t}), \right. \\ &\quad \left. \mathbf{K}_{\tilde{\mathbf{x}}\tilde{\mathbf{x}}}^{(t)} - \mathbf{K}_{\tilde{\mathbf{x}}\mathbf{w}}^{(t)} (\mathbf{K}_{\mathbf{w}\mathbf{w}}^{(t)})^{-1} \mathbf{K}_{\mathbf{w}\tilde{\mathbf{x}}}^{(t)}\right). \end{aligned} \quad (\text{B.10})$$

We now introduce a variational distribution over the inducing variables,

$$q_{\psi_t}(\mathbf{g}_t) = \mathcal{N}(\boldsymbol{\mu}_t, \mathbf{S}_t),$$

and define the sparse variational family by

$$q_{\psi_t, \gamma_t}(\mathbf{f}_t, \mathbf{g}_t \mid \tilde{\mathbf{x}}_t) = p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) q_{\psi_t}(\mathbf{g}_t).$$

Marginalizing over $\mathbf{g}_t$ gives the implied variational distribution over the latent function values,

$$q_{\psi_t, \gamma_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) = \int p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) q_{\psi_t}(\mathbf{g}_t) d\mathbf{g}_t.$$

Using this variational family, the log-likelihood term at time t can be written as

$$\begin{aligned} \log p_{\gamma_t}(\mathbf{a}_t \mid \tilde{\mathbf{x}}_t) &= \log \iint p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t) d\mathbf{f}_t d\mathbf{g}_t \\ &= \log \iint q_{\psi_t}(\mathbf{g}_t) p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) \frac{p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t)}{q_{\psi_t}(\mathbf{g}_t)} d\mathbf{f}_t d\mathbf{g}_t \\ &\geq \iint q_{\psi_t}(\mathbf{g}_t) p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) \log \left(\frac{p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t) p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t)}{q_{\psi_t}(\mathbf{g}_t)} \right) d\mathbf{f}_t d\mathbf{g}_t \\ &= \mathbb{E}_{q_{\psi_t, \gamma_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t)} [\log p_{\gamma_t}(\mathbf{a}_t \mid \mathbf{f}_t)] - \text{KL}(q_{\psi_t}(\mathbf{g}_t) \parallel p_{\gamma_t}(\mathbf{g}_t \mid \mathbf{w}_t)). \end{aligned} \quad (\text{B.11})$$

The last equality follows by marginalizing over $\mathbf{g}_t$ to obtain

$$q_{\psi_t, \gamma_t}(\mathbf{f}_t \mid \tilde{\mathbf{x}}_t) = \int p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) q_{\psi_t}(\mathbf{g}_t) d\mathbf{g}_t,$$

and by using

$$\int p_{\gamma_t}(\mathbf{f}_t \mid \mathbf{g}_t, \tilde{\mathbf{x}}_t, \mathbf{w}_t) d\mathbf{f}_t = 1.$$

Substituting the bound in Eq. (B.11) into Eq. (B.6), we obtain

$$\begin{aligned} \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) &\geq \sum_{t=1}^T \mathbb{E}_{q_{\boldsymbol{\psi}}(\mathbf{z}|\mathbf{x})} \left[\mathbb{E}_{q_{\boldsymbol{\psi}_t, \boldsymbol{\gamma}_t}(\mathbf{f}_t|\tilde{\mathbf{x}}_t)} [\log p_{\boldsymbol{\gamma}_t}(\mathbf{a}_t | \mathbf{f}_t)] \right] \\ &\quad - \sum_{t=1}^T \text{KL}(q_{\boldsymbol{\psi}_t}(\mathbf{g}_t) \parallel p_{\boldsymbol{\gamma}_t}(\mathbf{g}_t | \mathbf{w}_t)) - \text{KL}(q_{\boldsymbol{\psi}}(\mathbf{z} | \mathbf{x}) \parallel p_{\boldsymbol{\gamma}}(\mathbf{z} | \mathbf{x})). \end{aligned} \quad (\text{B.12})$$

Define

$$\begin{aligned} \mathcal{L}_{\text{ind}}(\boldsymbol{\psi}, \boldsymbol{\gamma}) &:= \sum_{t=1}^T \mathbb{E}_{q_{\boldsymbol{\psi}}(\mathbf{z}|\mathbf{x})} \left[\mathbb{E}_{q_{\boldsymbol{\psi}_t, \boldsymbol{\gamma}_t}(\mathbf{f}_t|\tilde{\mathbf{x}}_t)} [\log p_{\boldsymbol{\gamma}_t}(\mathbf{a}_t | \mathbf{f}_t)] \right] \\ &\quad - \sum_{t=1}^T \text{KL}(q_{\boldsymbol{\psi}_t}(\mathbf{g}_t) \parallel p_{\boldsymbol{\gamma}_t}(\mathbf{g}_t | \mathbf{w}_t)) - \text{KL}(q_{\boldsymbol{\psi}}(\mathbf{z} | \mathbf{x}) \parallel p_{\boldsymbol{\gamma}}(\mathbf{z} | \mathbf{x})). \end{aligned} \quad (\text{B.13})$$

Then

$$\log p_{\boldsymbol{\gamma}}(\mathbf{a} | \mathbf{x}) \geq \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\gamma}) \geq \mathcal{L}_{\text{ind}}(\boldsymbol{\psi}, \boldsymbol{\gamma}). \quad (\text{B.14})$$

Hence, $\mathcal{L}_{\text{ind}}(\boldsymbol{\psi}, \boldsymbol{\gamma})$ is a valid lower bound on the log-evidence and can be used as the optimization objective for sparse variational inference in the SeqGPLVM. Equivalently, in practice one minimizes the negative inducing-point ELBO,

$$\mathcal{J}_{\text{SeqGPLVM}} = -\mathcal{L}_{\text{ind}}(\boldsymbol{\psi}, \boldsymbol{\gamma}).$$

Compared with the bound in Eq. (B.8), this bound offers two key advantages. The expected log-likelihood is computed with respect to the sparse marginal posterior $q_{\boldsymbol{\psi}_t, \boldsymbol{\gamma}_t}(\mathbf{f}_t | \tilde{\mathbf{x}}_t)$, induced by the lower-dimensional variational distribution $q_{\boldsymbol{\psi}_t}(\mathbf{u}_t)$, rather than the full variational distribution $q_{\boldsymbol{\psi}_t}(\mathbf{f}_t | \tilde{\mathbf{x}}_t)$. In addition, the KL term in Eq. (B.12) depends only on the inducing variables, not on the full vector of latent function values. This substantially reduces the computational burden at each time point, since the main matrix inversion is of size $M_t \times M_t$ instead of $N_t \times N_t$, yielding a complexity of $O(M_t^3)$ rather than $O(N_t^3)$.

The inducing inputs can be optimized as variational parameters. However, it is a common practice to fix the inducing inputs to a subset of the observed augmented inputs, which further reduces the computational cost and stabilizes the optimization. The prior over the latent variable $\mathbf{Z}$ is typically chosen to be independent of the observed covariates and to be an i.i.d. Gaussian distribution,

$$p(\mathbf{z}) = \prod_{i=1}^N \mathcal{N}(\mathbf{z}_i | \mathbf{0}, I_J).$$

Also the variational distribution over the latent variable $\mathbf{Z}$ is typically chosen to

be a mean-field Gaussian distribution, both with respect to the data points and the dimensions of the latent variable,

$$q_\psi(\mathbf{z} \mid \mathbf{x}) = \prod_{i=1}^N \prod_{j=1}^J \mathcal{N}(z_{i,j} \mid \mu_{i,j}, \sigma_{i,j}^2),$$

where $\mu_{i,j}$ and $\sigma_{i,j}^2$ are the mean and variance parameters for the (j) -th dimension of the i -th latent variable, respectively. This is called the non-amortized variational inference setting, since the variational distribution over the latent variables does not depend on the observed covariates. The downside of this choice is that for the new data points, we need to repeat the optimization to find the corresponding mean and variance parameters, while freezing the fitted Gaussian process hyperparameters, which can be computationally expensive.

The number of inducing points M_t is a key hyperparameter that governs the trade-off between computational efficiency and approximation quality. There is no universally optimal choice for M_t , and it must generally be tuned to the dataset and available computational resources. In our experiments, at each time point t , the inducing set in the joint covariate–latent space is constructed as the Cartesian product of $\lfloor \sqrt{N_t} \rfloor$ inducing locations in the observed covariate space and 10 inducing locations in the latent space, where N_t denotes the number of observations at time point t . The covariate inducing locations are selected from the observed non-missing covariate vectors by greedy farthest-point sampling, that is, points are added sequentially so as to maximize their minimum distance from those already selected, yielding a well-dispersed subset of the empirical covariate space.

The latent inducing locations are chosen separately from the observation-specific latent variables and are initialized directly in the latent space using Latin hypercube sampling over a bounded interval. In this way, the latent space is partitioned into equally sized strata and one point is drawn from each stratum, yielding inducing locations that are well dispersed across the latent space.

C Additional Simulation Results

Figures C.1–C.3 report the results of the simulation study for the four-covariate case. The additional results for the four-covariate case lead to the same overall conclusions as in the two-covariate setting. Figure C.1 reports the bias, Figure C.2 reports the standard error, and Figure C.3 reports the coverage. Within each figure, the first two rows correspond to the weak unmeasured confounding setting ($c_U = 1$), and the last two rows correspond to the strong unmeasured confounding setting ($c_U = 2$). The first and third rows correspond to $\tau_F = 1$, while the second and fourth rows correspond to $\tau_C = 0.3$. The columns correspond to the N to T ratio, with panels (A), (D), (G), and (J) showing $\rho = 5$, panels (B), (E), (H), and (K) showing $\rho = 10$, and panels (C), (F), (I), and (L) showing $\rho = 50$. In particular, both **FE-MSM** and **SeqGPLVM-MSM** still improve over the naive **IPTW** estimator, while **FE-MSM** remains uniformly closer to **IPTW-True** and exhibits smaller estimated standard errors than **SeqGPLVM-MSM**. However, increasing the number of observed covariates from $d_X = 2$ to $d_X = 4$ does not lead to a systematic finite-sample improvement. Across scenarios, the $d_X = 4$ results are generally associated with larger average standard errors (Figure C.2). Although coverage occasionally improves in the more difficult settings, especially under stronger unmeasured confounding, these improvements appear to be driven mainly by wider intervals rather than smaller bias (Figures C.1 and C.3, panels (G)–(L)). As in the main results, the most challenging regime remains the short-panel setting with $\rho = 50$, particularly for smaller sample sizes, where the performance of both adjusted estimators deteriorates especially for **SeqGPLVM-MSM** (Figures C.1–C.3, panels (C), (F), (I), and (L)).

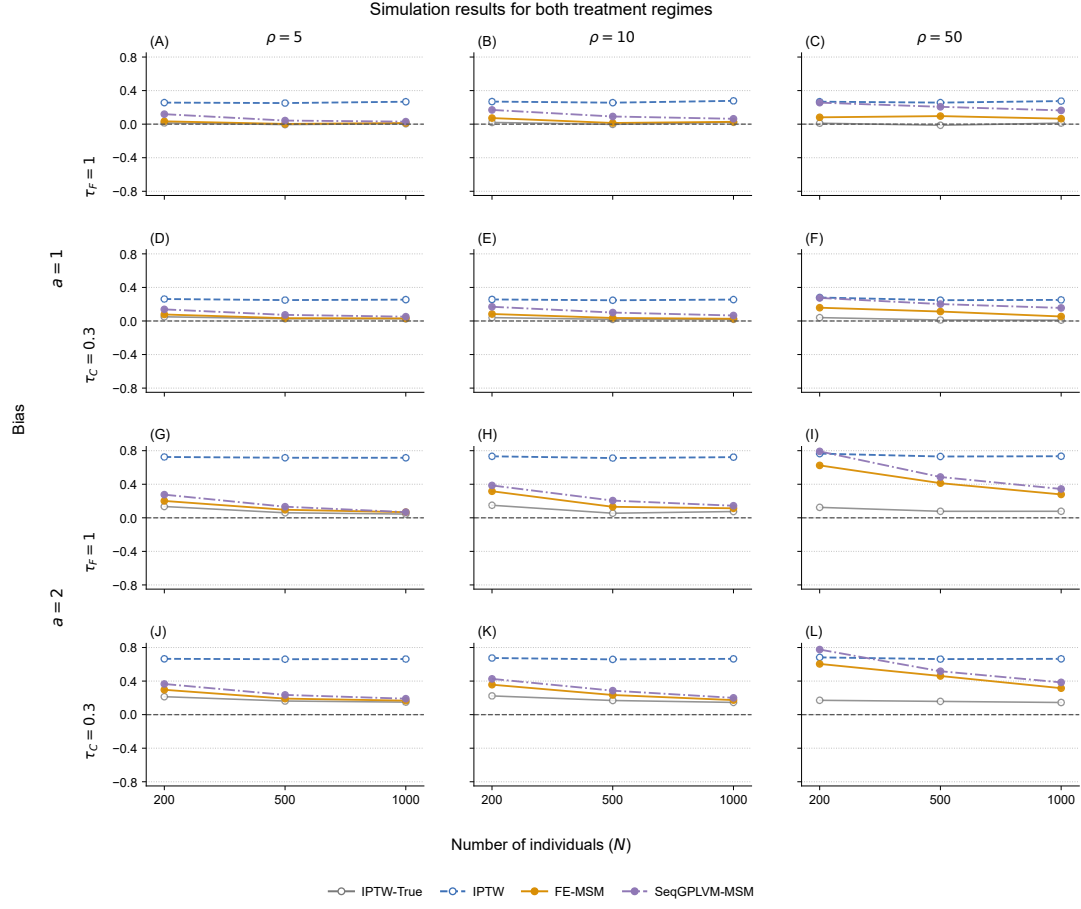


Figure C.1: Simulation bias for the four-covariate case. The panels report bias for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.

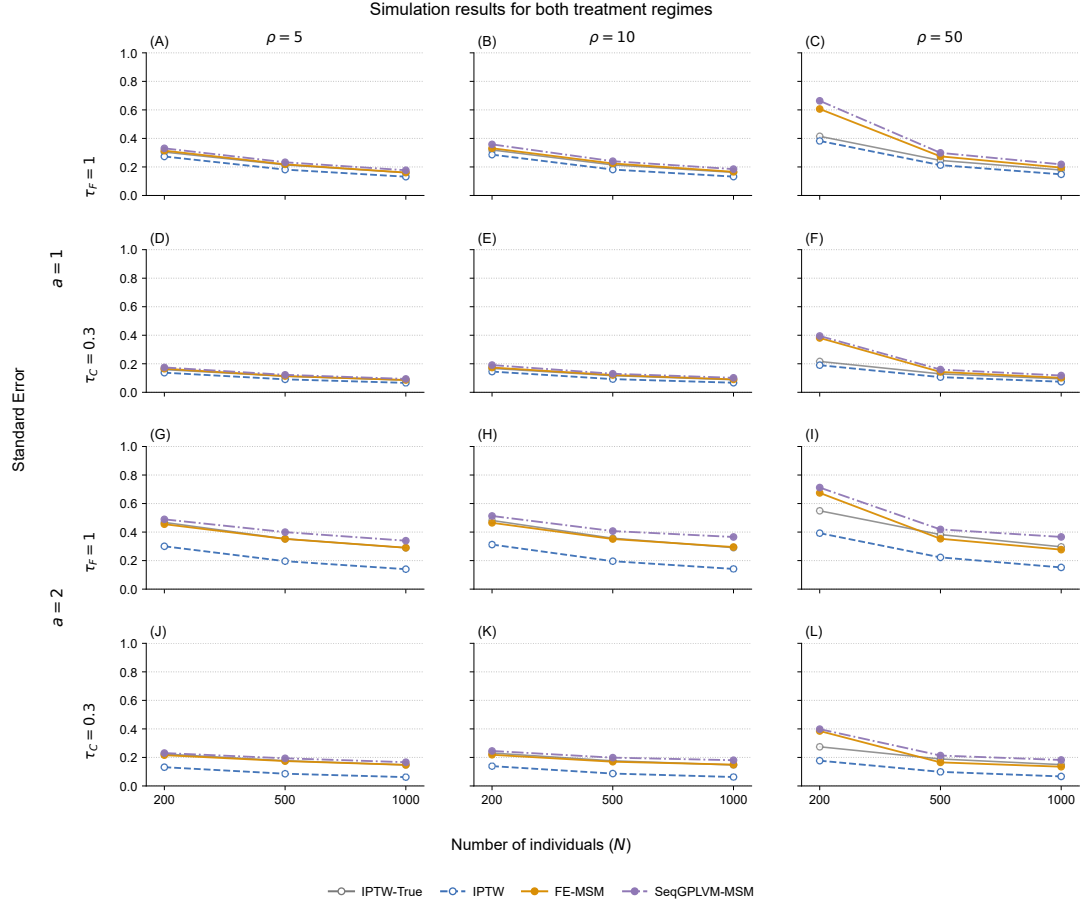


Figure C.2: Simulation standard error for the four-covariate case. The panels report standard error for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.

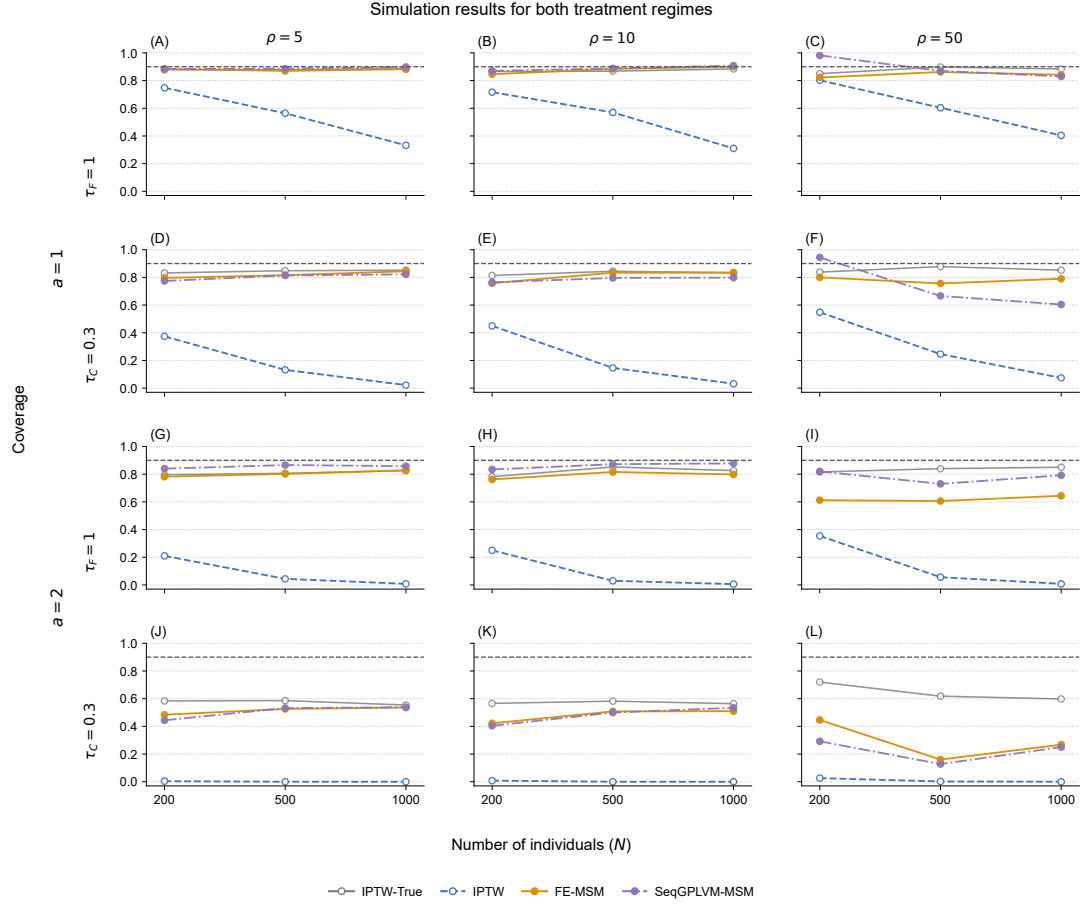


Figure C.3: Simulation coverage for the four-covariate case. The panels report coverage of the 90% confidence intervals for $c_U = 1$ and $c_U = 2$, for $\tau_F = 1$ and $\tau_C = 0.3$, across sample sizes N and ratios $\rho = N/T$.